

Zero-Shot Linear Quadratic Regulation from Reward-Free Data: Structured Successor Features with Stability and Finite-Sample Guarantees

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Abstract

Reinforcement learning (RL) and adaptive dynamic programming (ADP) algorithms for the linear quadratic regulator (LQR) learn the optimal gain for *one* performance index. Each time the designer changes the weights (Q, R) , these algorithms must re-collect or re-process data and re-run policy iteration. Zero-shot RL instead learns a representation from reward-free interaction and reads out the optimal policy for *any* reward revealed at test time, without further learning. Existing zero-shot methods, however, rely on neural approximation and give no stability or optimality guarantee. This paper brings zero-shot RL to linear optimal control. For linear systems with quadratic costs, we show three facts. First, the successor features of every linear policy are exact quadratic forms. Second, the map from (Q, R) to the Q-function kernel is linear, with a closed-form successor operator. Third, a fixed-size second-moment statistic of reward-free data is a sufficient zero-shot representation. On this basis, we propose a zero-shot LQR algorithm for discrete-time systems. An offline phase compresses arbitrary reward-free trajectories into the representation. A test phase maps any task (Q, R) , given explicitly or through reward labels, to a gain. The test phase uses either a data-free policy iteration or, with no iteration, successor-feature generalized policy improvement (GPI). We prove three guarantees. With noise-free persistently exciting data, the returned gains equal the Riccati-optimal gains for all tasks simultaneously. The GPI policy is exponentially stabilizing and no worse than any library policy. Under sub-Gaussian process noise, the following holds with probability $1 - \delta$ and uniformly over a compact task set: the gains are stabilizing, and their relative suboptimality is $\mathcal{O}(\sigma_w \sqrt{\log(1/\delta)/(N\lambda_{PE})})$. Here N is the number of samples and λ_{PE} is the persistency-of-excitation level. A certificate computed from data verifies stability and suboptimality for each task. The framework extends to continuous-time systems through an integral-data Gram representation and a data-free Kleinman iteration. On the unstable batch reactor, simulations confirm exactness to machine precision over 200 random tasks and the predicted sample complexity. The test-time cost per task stays at 0.4 ms for every data length, whereas per-task off-policy Q-learning takes 14–359 ms.

Keywords: Zero-shot reinforcement learning, Linear quadratic regulator, Successor features, Adaptive dynamic programming, Data-driven control, Persistency of excitation

1. Introduction

Reinforcement learning (RL) and adaptive dynamic programming (ADP) solve linear quadratic regulation (LQR) problems without knowledge of the system matrices [Lewis and Vrabie \(2009\)](#); [Kiumarsi et al. \(2018\)](#); [Jiang and Jiang \(2017\)](#); [Lewis et al. \(2012\)](#). For discrete-time (DT) systems $x_{k+1} = Ax_k + Bu_k$, on- and off-policy Q-learning [Bradtke et al. \(1994\)](#); [Bradtke and Barto \(1996\)](#); [Lagoudakis and Parr \(2003\)](#); [Kiumarsi et al. \(2014\)](#); [Rizvi and Lin \(2019\)](#); [Zhang et al. \(2023\)](#) implement Hewer’s policy iteration [Hewer \(1971\)](#) from measured state–input data. Integral RL and off-policy ADP [Vrabie et al. \(2009\)](#); [Jiang and Jiang \(2012\)](#); [Vamvoudakis \(2017\)](#); [Possieri and Sassano \(2022\)](#) implement Kleinman’s algorithm [Kleinman \(1968\)](#) for continuous-time (CT) systems $\dot{x} = Ax + Bu$. Two further lines of work characterize how much data such methods need: direct data-driven control [De Persis and Tesi \(2020\)](#); [van Waarde et al. \(2020\)](#); [Dörfler et al. \(2023\)](#); [Celi et al. \(2023\)](#), built on Willems’ fundamental lemma [Willems et al. \(2005\)](#), and learning-theoretic analyses of LQR [Dean et al.](#)

[\(2020\)](#); [Mania et al. \(2019\)](#); [Tu and Recht \(2019\)](#); [Krauth et al. \(2019\)](#); [Fazel et al. \(2018\)](#); [Abbasi-Yadkori and Szepesvári \(2011\)](#).

All of these methods solve the problem for *one fixed* performance index. The weights (Q, R) enter the Bellman equations that the methods solve from data. A designer changes the weights to trade tracking accuracy against actuator effort, to adapt to a new mission phase, or during routine controller tuning. After each change, an off-policy method must run policy iteration again on the stored data, and an on-policy method must collect new data. Engineers rarely know the weights in advance and tune them iteratively, so every tuning step costs a full learning run.

Machine-learning research studies the analogous question under the name *zero-shot RL* [Touati and Ollivier \(2021\)](#); [Touati et al. \(2023\)](#). An agent interacts with the environment *without any reward* and builds a representation. At test time, the agent must output a near-optimal policy for *any* reward function immediately, without further learning or planning in the environment. Successor features (SF) [Dayan \(1993\)](#); [Barreto](#)

et al. (2017); Borsa et al. (2019) form the first main family of methods. They assume rewards linear in known features and combine the SFs of several policies through generalized policy improvement (GPI). Forward-backward (FB) representations Touati and Ollivier (2021); Blier et al. (2021); Touati et al. (2023) form the second family. They learn a low-rank factorization $M^{\pi_z}(s, a, s') \approx F(s, a, z)^\top B(s')$ of the successor measure, i.e. of the expected future visits of a policy to each state, together with a task-conditioned policy family π_z . Follow-up work improves representation learning Park et al. (2024) and robustness to low-quality data Jeon et al. (2024). These methods rely on neural network approximation, and their evaluations use simulated locomotion and navigation benchmarks. They provide neither a closed-loop stability guarantee nor an error bound in terms of the amount and quality of data, and control engineering requires both before it accepts a controller.

Control research has also studied multitask and meta-learning formulations of LQR. Multitask LQG control Toso et al. (2026) learns one shared stabilizing controller across a distribution of systems and objectives. Its generalization bounds grow with a heterogeneity measure between tasks, defined through a bisimulation function. Because all tasks share the controller, the controller is biased away from each task-specific optimum. Meta-learning approaches such as MAML-LQR Finn et al. (2017); Toso et al. (2024) instead learn an initialization. A new task then reaches its own gain through further policy-gradient adaptation, which still requires online learning on that task. The task-space-complexity framework for zero-shot generalization in Cho et al. (2026) uses LQR as one of its running examples. It measures transfer through a performance-based divergence between source and target contexts and gives no closed-form policy map. None of these methods returns the exact gain for every task in a continuum from a single reward-free dataset, without further interaction or adaptation. These methods also give no stability certificate for such a gain. This paper provides both.

We call a method zero-shot in the representation-based sense if a reward-free representation yields a near-optimal policy for *any* task through a closed-form map, with provable stability and suboptimality guarantees. To the authors' knowledge, zero-shot RL in this sense has not been formulated or analyzed for the LQR problem. The ADP and data-driven LQR methods above also leave learning unamortized across performance indices. This paper connects the two lines of work. Our starting observation is that the linear-quadratic structure makes the objects that zero-shot RL approximates *exactly* finite-dimensional. With the quadratic features $\phi(z) = z \otimes z$, $z = [x^\top, u^\top]^\top$, every stage cost $x^\top Qx + u^\top Ru$ is linear in ϕ . The successor features of any linear policy are then quadratic forms, and the map from the task to the Q-function kernel of the policy is a linear operator in closed form. The contributions are as follows.

1. **Problem and exact representation.** We formulate the zero-shot LQR problem (Problem 4). We show that the successor features of LQR admit an exact forward-backward factorization, with a closed-form successor operator $\mathcal{L}_K$ that is linear in the task (Proposition 6). A

second-moment statistic of reward-free data suffices to evaluate every policy on every task, and its size depends on neither the number of samples nor the task.

2. **Algorithms.** For DT systems, we propose a two-phase algorithm. The offline phase compresses arbitrary reward-free, off-policy trajectories (Algorithm 1). At test time, the algorithm maps any task, given as weights or as reward labels, to a gain (Algorithm 2). It uses either a data-free policy iteration or, with no iteration, SF-GPI over a policy library. For CT systems, an integral-data Gram representation supports a data-free Kleinman iteration (Algorithm 3).
3. **Guarantees.** Let the data be noise-free and persistently exciting (PE, Definition 3). Then the returned gains equal the Riccati solutions for all tasks simultaneously, and every iterate is stabilizing (Theorem 8). These iterates coincide with those of per-task off-policy Q-learning, which needs quadratic PE where our method needs only linear PE. The SF-GPI policy is exponentially stabilizing and never worse than the best library policy (Theorem 9). Under sub-Gaussian process noise, the gains are stabilizing with probability $1 - \delta$. Their relative suboptimality is bounded uniformly over a compact task set by a quantity of order $\sigma_w \sqrt{\log(1/\delta)/(N\lambda_{\text{PE}})}$. A certificate computed from data verifies both properties for each task (Theorem 12). Theorem 15 gives the CT counterpart.
4. **Evidence.** We test the method on the open-loop unstable batch reactor. The experiments confirm exactness over 200 random tasks and a test-time cost independent of the data length. They also confirm the predicted $1/N$ decay of suboptimality for three PE levels, the near-optimality of zero-iteration SF-GPI, the CT extension, and task inference from noisy reward labels.

The structure has one more consequence, which we state explicitly. For LQR, the zero-shot representation is equivalent to the least-squares estimate of the one-step transition operator $\Theta = [A \ B]$. The test phase can therefore be read as certainty-equivalent control computed on demand, i.e. control designed for the estimated model as if that model were exact. This equivalence is informative: in the linear-quadratic case, the objects that FB and SF methods learn with neural networks reduce to closed forms whose error can be bounded. It also yields, to our knowledge, the first finite-sample, closed-loop guarantees for zero-shot RL on continuous state and action spaces. Four contributions go beyond certainty equivalence: the operator viewpoint (Proposition 6), the zero-iteration SF-GPI controller with its stability proof, the uniform-over-tasks guarantee, and the per-task certificate computed from data. The uniform guarantee follows from a *single* probabilistic event and needs no union bound over tasks.

Organization. Section 2 states the problem, and Section 3 derives the structured successor features. The DT algorithm and its analysis follow in Section 4, the CT extension in Section 5, and the simulations in Section 6. Section 7 discusses limitations, and the Appendix contains all proofs.

Notation. $A > 0$ (≥ 0) denotes a positive (semi)definite matrix, and $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the extreme eigenvalues

of a symmetric matrix. $\|\cdot\|$ is the Euclidean norm or the induced spectral norm, $\rho(\cdot)$ the spectral radius, and $\otimes$ the Kronecker product. $\text{vec}(\cdot)$ stacks columns, so that $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$ and $z^\top Wz = \text{vec}(W)^\top (z \otimes z)$. A matrix M is Schur if $\rho(M) < 1$ and Hurwitz if all its eigenvalues have negative real part. $\text{blkdiag}(\cdot)$ builds a block-diagonal matrix, and I_n is the $n \times n$ identity.

2. Problem Formulation

2.1. System, Tasks, and Costs

Consider the DT linear system

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + w_k = \Theta z_k + w_k, \\ z_k &:= \begin{bmatrix} x_k \\ u_k \end{bmatrix} \in \mathbb{R}^d, \end{aligned} \quad (1)$$

with state $x_k \in \mathbb{R}^n$, input $u_k \in \mathbb{R}^m$, $d := n + m$, unknown $\Theta := [A \ B] \in \mathbb{R}^{n \times d}$, and process noise w_k (possibly zero). A task is a weight pair (Q, R) collected in

$$\begin{aligned} W &:= \text{blkdiag}(Q, R) \in \mathbb{R}^{d \times d}, \\ c(W) &:= \lambda_{\min}(W) = \min\{\lambda_{\min}(Q), \lambda_{\min}(R)\}. \end{aligned} \quad (2)$$

For a linear policy $u = -Kx$ let $\Pi_K := [I_n \ -K^\top]^\top \in \mathbb{R}^{d \times n}$, so that the closed-loop matrix is $A - BK = \Theta \Pi_K$. The set of stabilizing gains is $\mathcal{K}(\Theta) := \{K : \rho(\Theta \Pi_K) < 1\}$. For $K \in \mathcal{K}(\Theta)$ the value matrix $P_K(W)$ is the unique solution of the Lyapunov equation

$$\begin{aligned} P_K &= (A - BK)^\top P_K (A - BK) + Q + K^\top R K \\ &= (\Theta \Pi_K)^\top P_K (\Theta \Pi_K) + \Pi_K^\top W \Pi_K, \end{aligned} \quad (3)$$

and $x_0^\top P_K x_0 = \sum_{t \geq 0} (x_t^\top Q x_t + u_t^\top R u_t)$ along the noise-free closed loop. We measure performance by

$$J_W(K) := \text{tr } P_K(W), \quad (4)$$

the expected cost from $x_0 \sim \mathcal{N}(0, I_n)$, which also equals the average cost under unit-covariance process noise. The optimal value $P^*(W)$ is the stabilizing solution of the discrete algebraic Riccati equation (DARE)

$$\begin{aligned} P &= A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A \\ &\quad + Q, \end{aligned} \quad (5)$$

with optimal gain $K^*(W) = (R + B^\top P^* B)^{-1} B^\top P^* A$ and $J_W^* = \text{tr } P^*(W)$. The associated Q-function kernel of a policy K is

$$\begin{aligned} H_K(W) &:= W + \Theta^\top P_K(W) \Theta = \begin{bmatrix} H^{xx} & H^{xu} \\ H^{ux} & H^{uu} \end{bmatrix}, \\ Q_K^W(x, u) &= z^\top H_K(W) z. \end{aligned} \quad (6)$$

2.2. Zero-Shot LQR

Assumption 1 (Tasks). (A, B) is stabilizable. The task set $\mathcal{W}$ is a compact set of weights $W = \text{blkdiag}(Q, R)$ with $Q > 0, R > 0$, and $c_{\mathcal{W}} := \min_{W \in \mathcal{W}} c(W) > 0$.

Assumption 2 (Reward-Free Data). The system (1) generates a dataset $\mathcal{D} = \{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$, possibly as several concatenated episodes. An arbitrary causal exploration rule chooses the inputs, for example open-loop excitation, any behavior policy, or a mixture. The learner observes no cost or reward. Let $\mathcal{F}_k$ be the σ -algebra generated by $z_0, \dots, z_k, w_0, \dots, w_{k-1}$. Then w_k is conditionally zero-mean given $\mathcal{F}_k$, and each coordinate of w_k is conditionally σ_w -sub-Gaussian.

Define the data matrices $Z := [z_0, \dots, z_{N-1}] \in \mathbb{R}^{d \times N}$ and $X^+ := [x_1, \dots, x_N] \in \mathbb{R}^{n \times N}$ and the empirical second moments

$$\hat{\Sigma}_{zz} := \frac{1}{N} Z Z^\top, \quad \hat{\Sigma}_{x^+z} := \frac{1}{N} X^+ Z^\top. \quad (7)$$

Definition 3 (PE Level). The data are persistently exciting (PE) if $\text{rank } Z = d$, equivalently $\hat{\Sigma}_{zz} > 0$. The PE level is $\lambda_{\text{PE}} := \lambda_{\min}(\hat{\Sigma}_{zz})$.

Problem 4 (Zero-Shot LQR). *Offline phase*: from $\mathcal{D}$ alone, compute a representation $\mathcal{R} = \mathcal{R}(\mathcal{D})$ whose size is independent of N and of any task. *Test phase*: a task $W \in \mathcal{W}$ is revealed, either explicitly or through reward labels $r_k = z_k^\top W z_k$ (possibly noisy) on $\mathcal{D}$. Return a gain $\hat{K}(W) = \Gamma(\mathcal{R}, W)$ without interacting with the system and without reprocessing $\mathcal{D}$; the only exception is one linear pass to infer W when only labels are given. The gain must satisfy:

- (i) $\hat{K}(W) \in \mathcal{K}(\Theta)$ (closed-loop stability);
- (ii) $J_W(\hat{K}(W)) \leq (1 + \eta_N) J_W^*$ with $\sup_{W \in \mathcal{W}} \eta_N \rightarrow 0$ as $N \rightarrow \infty$ (uniform near-optimality);
- (iii) the computational cost of Γ is independent of N .

Classical off-policy Q-learning [Bradtke et al. \(1994\)](#); [Kiumarsi et al. \(2014\)](#) satisfies (i)–(ii) for one task but violates (iii). Every task requires a new policy iteration, and each policy-evaluation step in it solves a least-squares problem over all N samples. On-policy methods also violate the offline requirement, because each task requires new interaction.

3. Structured Successor Features for LQR

Zero-shot RL rests on the successor representation [Dayan \(1993\)](#). For a policy π , the successor features with respect to a feature map ϕ are $\psi^\pi(z) = \sum_{t \geq 0} \phi(z_t)$ along the trajectory started at $z_0 = z$ [Barreto et al. \(2017\)](#). If the reward is linear in the features, $r = \phi^\top w$, then $Q_r^\pi = \psi^\pi{}^\top w$ for every w . Evaluating a known policy on a new task then reduces to an inner product. FB representations [Touati and Ollivier \(2021\)](#) go one step further. They learn a family of policies π_ζ together with a factorization $Q_r^{\pi_\zeta}(z) = F(z, \zeta)^\top \zeta$, where a *backward* map computes the task code $\zeta = \mathbb{E}[B(z)r(z)]$. At test time, the method reads out the policy π_ζ directly.

For LQR we use the quadratic features and task code

$$\phi(z) := z \otimes z \in \mathbb{R}^{d^2}, \quad w := \text{vec}(W), \quad z^\top W z = \phi(z)^\top w. \quad (8)$$

Define the state–action transition matrix of the noise-free closed loop under K ,

$$G_K := \Pi_K \Theta \in \mathbb{R}^{d \times d}, \quad z_{t+1} = G_K z_t \text{ when } u_t = -K x_t, \ t \geq 1. \quad (9)$$

Lemma 5. For every K , the nonzero eigenvalues of G_K coincide with those of $A - BK$, so $\rho(G_K) = \rho(A - BK)$. For $K \in \mathcal{K}(\Theta)$, the kernel (6) is the unique solution of

$$H_K(W) = W + G_K^\top H_K(W) G_K, \quad (10)$$

and $P_K(W) = \Pi_K^\top H_K(W) \Pi_K$.

Proposition 6 (Exact Successor Features and Successor Operator). Let $K \in \mathcal{K}(\Theta)$ and consider the noise-free closed loop. Then:

- (a) the successor features of K are $\psi^K(z) = (I_{d^2} - G_K \otimes G_K)^{-1}(z \otimes z)$, a quadratic form in z ;
- (b) for every task, $Q_K^W(z) = \psi^K(z)^\top \text{vec}(W)$ and $\text{vec}(H_K(W)) = \mathcal{L}_K \text{vec}(W)$ with the successor operator

$$\mathcal{L}_K := (I_{d^2} - G_K^\top \otimes G_K^\top)^{-1}; \quad (11)$$

- (c) G_K , and hence $\mathcal{L}_K$ and ψ^K , depend on the data-generating system only through Θ .

Proposition 6 states that in LQR the Q-function of any linear policy is *linear in the task*, with an explicit operator. The result maps onto the FB template. The backward map is the least-squares projection of rewards on ϕ ,

$$\hat{w} = \left(\sum_k \phi(z_k) \phi(z_k)^\top \right)^\dagger \sum_k \phi(z_k) r_k, \quad (12)$$

which recovers w exactly from noise-free labels whenever the quadratic features of the data span the symmetric matrices. The forward map is $F(z, W) = \psi^{K^*(W)}(z)$, so that $Q_W^*(z) = F(z, W)^\top \text{vec}(W)$. Generic FB approximates both maps with neural networks. In LQR, neither map needs approximation: ψ^K has a closed form once Θ is known, and policy iteration reaches $K^*(W)$ with every evaluation given by (10). Proposition 6(c) identifies what the offline phase must learn: the one-step transition operator Θ . When the data are PE,

$$\hat{\Theta} := \hat{\Sigma}_{x^+z} \hat{\Sigma}_{zz}^{-1} = X^+ Z^\top (Z Z^\top)^{-1} \quad (13)$$

is its least-squares estimate. The pair $\mathcal{R} := (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x^+z})$ has $d^2 + nd$ entries for every N , so it is an admissible zero-shot representation. We write $\hat{G}_K := \Pi_K \hat{\Theta}$, $\hat{\mathcal{L}}_K$, $\hat{H}_K(W)$ and $\hat{P}_K(W)$ for the quantities computed with $\hat{\Theta}$ in place of Θ .

Remark 7 (Why Linear Rather Than Quadratic Regression). Off-policy Q-learning Bradtke et al. (1994); Kiumarsi et al. (2014) evaluates K by regressing $z_k^\top W z_k$ on $\phi(z_k) - \phi(\Pi_K x_{k+1})$. This regression needs the quadratic PE condition $\text{rank}[\phi(z_0), \dots, \phi(z_{N-1})] = d(d+1)/2$, and therefore at least $d(d+1)/2$ samples. Estimating the linear operator Θ needs only $\text{rank } Z = d$. Under process noise, the term $\text{tr}(H \text{Cov}(w))$ also biases the quadratic regression, whereas the estimate (13) is consistent.

Algorithm 1 Offline Phase: Reward-Free Representation (DT)

Input: reward-free data $\{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$; optional anchor tasks $W_1, \dots, W_M$

- 1: $\hat{\Sigma}_{zz} \leftarrow \frac{1}{N} \sum_k z_k z_k^\top$, $\hat{\Sigma}_{x^+z} \leftarrow \frac{1}{N} \sum_k x_{k+1} z_k^\top$ ▷ streaming, $O(d^2)$ memory
- 2: check PE: $\lambda_{\text{PE}} = \lambda_{\min}(\hat{\Sigma}_{zz}) > 0$; $\hat{\Theta} \leftarrow \hat{\Sigma}_{x^+z} \hat{\Sigma}_{zz}^{-1}$
- 3: **for** $i = 1, \dots, M$ **do** ▷ optional policy library
- 4: $K_i \leftarrow \text{Mode PI of Algorithm 2 (or DARE of } \hat{\Theta}) \text{ for } W_i$
- 5: $\hat{\mathcal{L}}_{K_i} \leftarrow (I - \hat{G}_{K_i}^\top \otimes \hat{G}_{K_i}^\top)^{-1}$
- 6: **return** $\mathcal{R} = (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x^+z})$ and library $\{(K_i, \hat{\mathcal{L}}_{K_i})\}_{i=1}^M$; the raw data may be discarded

4. Zero-Shot LQR for Discrete-Time Systems

4.1. Algorithms

The offline phase (Algorithm 1) accumulates the second moments. Optionally, it also precomputes a *policy library*: M gains $K_1, \dots, K_M$ that stabilize $\hat{\Theta}$ (for instance $\hat{K}(W_i)$ for anchor tasks $W_1, \dots, W_M$), together with their successor operators $\hat{\mathcal{L}}_{K_i}$. The test phase (Algorithm 2) supports two modes.

Mode PI (policy iteration in representation space). Starting from a gain $K_0 \in \mathcal{K}(\hat{\Theta})$, iterate

$$\hat{H}_j = W + \hat{G}_{K_j}^\top \hat{H}_j \hat{G}_{K_j}, \quad K_{j+1} = (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}. \quad (14)$$

Each step solves a $d \times d$ Lyapunov equation and touches no data. The library can supply the initial gain $K_0 = \arg \min_{K_i} \text{tr}(\Pi_{K_i}^\top \hat{H}_{K_i}(W) \Pi_{K_i})$ at the cost of M matrix–vector products with the precomputed $\hat{\mathcal{L}}_{K_i}$. The method then needs no task-specific initial stabilizing gain.

Mode GPI (no iteration). For each library policy, compute $\hat{H}_i = \hat{H}_{K_i}(W)$ from $\text{vec}(\hat{H}_i) = \hat{\mathcal{L}}_{K_i} \text{vec}(W)$, the improved gain $K_i^+ = (\hat{H}_i^{uu})^{-1} \hat{H}_i^{ux}$, and the value kernel $S_i = \hat{H}_i^{xx} - \hat{H}_i^{xu} K_i^+$. Then apply the switched linear policy

$$\pi_W^{\text{GPI}}(x) = -K_{i^*}^+ x, \quad i^*(x) \in \arg \min_i x^\top S_i x. \quad (15)$$

This policy is exactly GPI Barreto et al. (2017), $\pi(x) \in \arg \min_u \min_i Q_{K_i}^W(x, u)$, written in closed form. Its test-time cost is M products of a $d^2 \times d^2$ matrix with a vector.

4.2. Exact Zero-Shot Optimality with Noise-Free Data

Theorem 8 (Exactness). Let $w_k \equiv 0$, let the data be PE, and let (A, B) be stabilizable. Then $\hat{\Theta} = \Theta$, and for every W with $Q > 0$, $R > 0$ and every $K_0 \in \mathcal{K}(\Theta)$ the iteration (14) satisfies:

- (a) $K_j \in \mathcal{K}(\Theta)$ for all j ;
- (b) $P^*(W) \leq P_{K_{j+1}}(W) \leq P_{K_j}(W)$;
- (c) $K_j \rightarrow K^*(W)$ and $P_{K_j}(W) \rightarrow P^*(W)$ quadratically.

If, in addition, the quadratic features of the data span the symmetric matrices, then the iterates coincide with those of per-task off-policy Q-learning Bradtke et al. (1994); Kiumarsi et al. (2014) run on $\mathcal{D}$ with the same W and K_0 . In that case, the backward map (12) also recovers W exactly from noise-free labels.

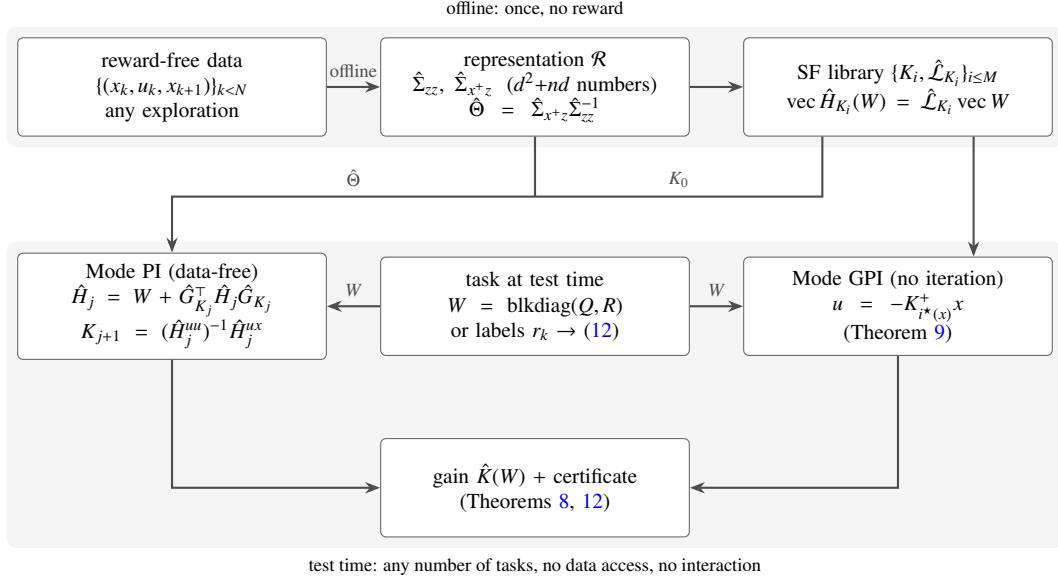


Figure 1: Zero-shot LQR. The offline phase compresses reward-free data into a fixed-size representation and, optionally, an SF library. At test time, the method maps any task to a gain without touching the data or the plant.

Algorithm 2 Test Phase: Zero-Shot Gain for a New Task (DT)

Input: representation $\mathcal{R}$, library, task W or labels $\{r_k\}$; accuracy ϵ_{tol} ; optional noise level $\bar{\epsilon}$

- 1: **if** only labels are given **then**
- 2: $W \leftarrow$ backward map (12); project Q, R onto $\{\geq \epsilon I\}$
- 3: **for** $i = 1, \dots, M$ **do**
- 4: $\text{vec}(\hat{H}_i) \leftarrow \hat{L}_{K_i} \text{vec}(W)$
- 5: **if** Mode GPI **then**
- 6: **return** switched policy (15)
- 7: $K_0 \leftarrow \arg \min_i \text{tr}(\Pi_{K_i}^T \hat{H}_i \Pi_{K_i})$, $j \leftarrow 0$
- 8: **repeat**
- 9: $\hat{H}_j \leftarrow$ solve $\hat{H}_j = W + \hat{G}_{K_j}^T \hat{H}_j \hat{G}_{K_j}$; $K_{j+1} \leftarrow (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}$;
 $j \leftarrow j + 1$
- 10: **until** $\|K_j - K_{j-1}\| < \epsilon_{\text{tol}}$
- 11: $\hat{K} \leftarrow K_j$, $\hat{P} \leftarrow \Pi_{\hat{K}}^T \hat{H}_{\hat{K}} \Pi_{\hat{K}}$
- 12: **certificate:** if $\bar{\epsilon} \leq c(W)/(12\|\hat{P}\|)$, report “stabilizing,
 $J_W(\hat{K}) \leq (1 + 20\bar{\epsilon}\|\hat{P}\|/c(W))J_W^*$ ” ▷ Theorem 12
- 13: **return** $\hat{K}$

The theorem holds for all tasks at once: one reward-free dataset, compressed once, solves every LQR problem on the plant exactly. Only linear PE is needed, which requires $N \geq d$ samples; per-task Q-learning needs $N \geq d(d+1)/2$.

Theorem 9 (Zero-Iteration SF-GPI). *Let $\hat{\Theta} = \Theta$ (e.g., under the hypotheses of Theorem 8), let $K_1, \dots, K_M \in \mathcal{K}(\Theta)$, and let W satisfy $Q > 0, R > 0$. Define $V(x) := \min_i x^T S_i x$. Then along the closed loop $x_{t+1} = Ax_t + B\pi_W^{\text{GPI}}(x_t)$:*

- (a) $V(x_{t+1}) \leq V(x_t) - x_t^T Q x_t$, and $\lambda_{\min}(Q)\|x\|^2 \leq V(x) \leq \max_i \|S_i\| \|x\|^2$; hence the origin is globally exponentially stable, with $V(x_t) \leq (1 - \lambda_{\min}(Q)/\max_i \|S_i\|)^t V(x_0)$;
- (b) the accumulated cost satisfies $\sum_{t \geq 0} (x_t^T Q x_t + u_t^T R u_t) \leq$

$$V(x_0) \leq \min_i x_0^T P_{K_i}(W) x_0 \text{ for every } x_0.$$

For $M = 1$, SF-GPI reduces to one Hwerner improvement step from K_1 . For $M > 1$, it is no worse than the best library policy from every initial state, and it solves no equation at test time.

4.3. Finite-Sample Guarantees with Noisy Data

Let $\bar{z}^2 \geq \lambda_{\max}(\hat{\Sigma}_{zz})$ and $\lambda_0 \leq \lambda_{\text{PE}}$ be deterministic levels, and define

$$\bar{\epsilon}_N(\delta) := \sqrt{\frac{4n\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log(1 + \bar{z}^2/\lambda_0) + \log \frac{n}{\delta} \right)}. \quad (16)$$

Lemma 10 (Representation Error). *Under Assumption 2, for any $\delta \in (0, 1)$ the event*

$$\mathcal{E} := \{\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I\} \cap \{\|\hat{\Theta} - \Theta\| > \bar{\epsilon}_N(\delta)\}$$

has probability at most δ .

Lemma 11 (One-Sided Perturbation). *Let $\Theta_1, \Theta_2 \in \mathbb{R}^{n \times d}$ with $\|\Theta_1 - \Theta_2\| \leq \epsilon$, let W satisfy $c = c(W) > 0$, let K stabilize Θ_1 , and let P_1 be its value matrix on Θ_1 with $p_1 = \|P_1\|$. If $\epsilon \leq c/(6p_1)$, then K stabilizes Θ_2 , and its value matrix on Θ_2 satisfies $P_2 \leq (1 + 6\epsilon p_1/c)P_1$.*

Theorem 12 (Uniform Zero-Shot Guarantee and Certificate). *Let Assumptions 1–2 hold, fix $\delta \in (0, 1)$, and let $\hat{K}(W), \hat{P}(W)$ be the output of Algorithm 2 (Mode PI run to convergence) with $\bar{\epsilon} = \bar{\epsilon}_N(\delta)$. With probability at least $1 - \delta$, whenever $\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I$, the following hold simultaneously for all $W \in \mathcal{W}$.*

- (a) **Certificate.** *If $\bar{\epsilon}_N(\delta) \leq c(W)/(12\|\hat{P}(W)\|)$, then $\hat{K}(W) \in \mathcal{K}(\Theta)$ and*

$$P^*(W) \leq P_{\hat{K}}(W) \leq \left(1 + 20 \bar{\epsilon}_N \frac{\|\hat{P}\|}{c(W)}\right) P^*(W), \quad (17)$$

$$\|\hat{P} - P^*\| \leq 14 \bar{\epsilon}_N \frac{\|\hat{P}\|^2}{c(W)}, \quad \|\hat{K} - K^*\| \leq \frac{\|\hat{P}\|}{c(W)} \sqrt{30 \bar{\epsilon}_N}.$$

(b) Uniform rate. Let $\bar{p}_W := \max_{W \in \mathcal{W}} \|P^*(W)\|$. If $\bar{\varepsilon}_N(\delta) \leq c_W/(18\bar{p}_W)$, then every $W \in \mathcal{W}$ satisfies the certificate in (a), and

$$\sup_{W \in \mathcal{W}} \frac{J_W(\hat{K}(W)) - J_W^*}{J_W^*} \leq 27 \frac{\bar{p}_W}{c_W} \bar{\varepsilon}_N(\delta) \quad (18)$$

which is

$$O\left(\frac{\bar{p}_W}{c_W} \sigma_w \sqrt{\frac{n(d \log(1 + \bar{z}^2/\lambda_0) + \log(n/\delta))}{N \lambda_0}}\right). \quad (19)$$

Remark 13 (Role of Data and Excitation). The bound (18) decreases as $1/\sqrt{N\lambda_0}$: halving the error requires either four times more data or four times more excitation. Both λ_{PE} and $\bar{z}^2$ follow from $\hat{\Sigma}_{zz}$, so the bound and the certificate in (a) can be evaluated from the representation alone, given σ_w . The guarantee involves a single probabilistic event and needs no union bound over $\mathcal{W}$. It therefore holds for all tasks at once, which makes it a zero-shot guarantee.

Remark 14 (Tightness). The rate in (18) is linear in $\bar{\varepsilon}_N$, which suffices for an explicit certificate computed from data. For small model errors, local analyses of certainty-equivalent LQR [Mania et al. \(2019\)](#) show that the suboptimality is quadratic in the model error, i.e. $O(1/(N\lambda_{PE}))$. Our experiments (Fig. 3) show this faster rate. A uniform-over-tasks version of the quadratic rate with explicit constants remains open.

5. Extension to Continuous-Time Systems

Consider $\dot{x} = Ax + Bu$ with unknown (A, B) , cost $\int_0^\infty (x^\top Qx + u^\top Ru) dt$, and optimal gain $K^*(W) = R^{-1}B^\top P^*$, where P^* solves the continuous-time algebraic Riccati equation (CARE). Kleinman’s algorithm [Kleinman \(1968\)](#) iterates $(A - BK_j)^\top P_j + P_j(A - BK_j) + Q + K_j^\top RK_j = 0$, $K_{j+1} = R^{-1}B^\top P_j$. Following [Jiang and Jiang \(2012\)](#), we apply an exploratory input $u = -K_j x + e(t)$ on $[t_0, t_l]$. For a partition $t_0 < t_1 < \dots < t_l$, we form the rows below, where $\bar{x}$ denotes the vector with $x^\top Px = \bar{x}^\top \text{svec}(P)$:

$$\begin{aligned} \delta_{xx} &= [\bar{x}^\top]_{t_{l-1}}^{t_l}, \\ I_{xx} &= \left[\int_{t_{l-1}}^{t_l} (x \otimes x)^\top d\tau \right]_i, \\ I_{xu} &= \left[\int_{t_{l-1}}^{t_l} (x \otimes u)^\top d\tau \right]_i. \end{aligned}$$

For a gain K_j , the Kleinman step is equivalent to the linear equation [Jiang and Jiang \(2012\)](#)

$$\begin{aligned} & \underbrace{[\delta_{xx}, -2I_{xx}(I_n \otimes K_j^\top R) - 2I_{xu}(I_n \otimes R)]}_{=DT(K_j, R)} \begin{bmatrix} \text{svec}(P_j) \\ \text{vec}(K_{j+1}) \end{bmatrix} \\ &= \underbrace{-I_{xx} \text{vec}(Q + K_j^\top RK_j)}_{=Ds(K_j, W)}, \end{aligned} \quad (20)$$

where $D := [\delta_{xx}, I_{xx}, I_{xu}]$ is the raw data matrix, and $T(\cdot)$ and $s(\cdot)$ are known linear maps that depend on the task but not on

Algorithm 3 Zero-Shot LQR for CT Systems

Input: offline: integral data $(\delta_{xx}, I_{xx}, I_{xu})$; a gain K_0 that stabilizes (A, B)

- 1: **offline:** $\mathcal{R}_{CT} \leftarrow D^\top D$; discard the data
- 2: **test** (any W): $j \leftarrow 0$
- 3: **repeat**
- 4: $T \leftarrow T(K_j, R)$, $s \leftarrow s(K_j, W)$; solve $T^\top \mathcal{R}_{CT} T \theta = T^\top \mathcal{R}_{CT} s$; $K_{j+1} \leftarrow$ the K -block of θ ; $j \leftarrow j + 1$
- 5: **until** $\|K_j - K_{j-1}\| < \epsilon_{\text{tol}}$
- 6: **return** $\hat{K}(W) = K_j$

the data. Consequently, the normal equations of (20) depend on the data only through the Gram matrix

$$\mathcal{R}_{CT} := D^\top D \in \mathbb{R}^{p \times p}, \quad p = \frac{n(n+1)}{2} + n^2 + nm, \quad (21)$$

whose size is independent of the number of intervals l . Algorithm 3 is the resulting zero-shot method.

Theorem 15 (CT Exactness). Suppose the data are noise-free and satisfy the rank condition $\text{rank}[I_{xx}, I_{xu}] = n(n+1)/2 + nm$ of [Jiang and Jiang \(2012\)](#). Let K_0 be a single gain that makes $A - BK_0$ Hurwitz. Then, for every W with $Q > 0$, $R > 0$, the iterates of Algorithm 3 coincide with Kleinman’s iterates for W . Hence every K_j is stabilizing and $K_j \rightarrow K^*(W)$ quadratically. The test-time cost per iteration is $O(p^3)$, independent of l .

The CT method needs a known stabilizing K_0 because Kleinman’s iteration must start from one; the DT method obtains K_0 from the library instead. One K_0 serves all tasks, since the stability of $A - BK_0$ is independent of W .

6. Simulation Results

6.1. Setup

We use Rosenbrock’s batch reactor [Walsh et al. \(2002\)](#); [De Persis and Tesi \(2020\)](#),

$$\begin{aligned} A_c &= \begin{bmatrix} 1.38 & -0.2077 & 6.715 & -5.676 \\ -0.5814 & -4.29 & 0 & 0.675 \\ 1.067 & 4.273 & -6.654 & 5.893 \\ 0.048 & 4.273 & 1.343 & -2.104 \end{bmatrix}, \\ B_c &= \begin{bmatrix} 0 & 0 \\ 5.679 & 0 \\ 1.136 & -3.146 \\ 1.136 & 0 \end{bmatrix}, \end{aligned}$$

This system is open-loop unstable (eigenvalues 1.99, 0.064, -5.06, -8.67), with $n = 4$, $m = 2$, $d = 6$. The DT model is the zero-order-hold discretization with sampling time $h = 0.1$ s ($\rho(A) = 1.22$). Each DT dataset consists of episodes of 10 steps from $x_0 \sim \mathcal{N}(0, I)$ with open-loop inputs $u_k \sim \mathcal{N}(0, \sigma_e^2 I)$. The data collection uses no feedback, no stabilizing gain, and no reward. We draw test tasks as $Q = U \text{diag}(10^{a_i}) U^\top$, $R = V \text{diag}(10^{b_i}) V^\top$ with random orthogonal U, V and $a_i, b_i \sim \mathcal{U}(-1, 1)$. The weights therefore

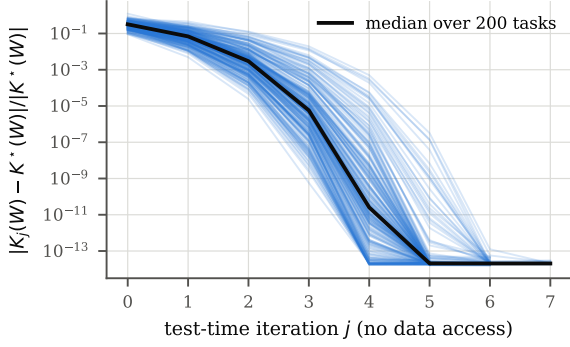


Figure 2: E1: relative gain error of the data-free test-time policy iteration for 200 random tasks from one reward-free dataset ($N = 200$). Thin lines: individual tasks; thick line: median.

Table 1: E1: per-task computation time (mean over 20 tasks) and the largest gap between the gains of the two methods.

N	offline (ms)	zero-shot (ms)	off-policy QL (ms)	max $\ \hat{K} - K_{QL}\ $
200	0.05	0.37	14.5	3.1×10^{-13}
1000	0.08	0.39	73.3	1.4×10^{-12}
5000	0.14	0.39	358.5	2.2×10^{-13}

span two orders of magnitude and are non-diagonal. We compute ground truth from the DARE and CARE of the true model. The accompanying repository contains the code that reproduces every number (code/experiments.py). [\[TODO: add repository link\]](#)

6.2. E1: Exactness and Test-Time Cost (DT, Noise-Free)

With $N = 200$ samples (20 episodes, $\sigma_e = 1$, $\lambda_{PE} = 0.138$), we ran Mode PI for 200 random tasks from the same initial gain. All iterates of all tasks were stabilizing. The largest relative gain error was 5.6×10^{-14} , i.e. machine precision, and convergence took 5.6 iterations on average (at most 7). Fig. 2 shows the quadratic convergence. Table 1 compares the per-task test-time cost with that of per-task off-policy Q-learning on the same data. The two methods return the same gains up to 1.4×10^{-12} . The cost of Q-learning grows linearly with N , while the cost of our method is independent of N .

6.3. E2: Sample Complexity and PE Level (DT, Noisy)

We added process noise with $\sigma_w = 0.05$ and varied the exploration level over $\sigma_e \in \{0.3, 1, 3\}$, which yields PE levels $\lambda_{PE} \approx 0.09, 0.15, 0.33$. For each N from 100 to 3×10^4 , we averaged five independent datasets and evaluated 100 tasks. Fig. 3(a) shows that $\|\hat{\Theta} - \Theta\|$ decays as $N^{-1/2}$ and decreases as the PE level grows, as Lemma 10 predicts. The high-probability bound $\bar{\epsilon}_N(\delta)$ with $\delta = 0.05$ lies above the average observed error at every N , by a factor of about 6–10. Fig. 3(b) reports the worst-task and median-task relative suboptimality. All $3 \times 6 \times 5 \times 100 = 9000$ returned gains stabilized the true plant. For $\sigma_e = 0.3, 1, 3$, the worst-task suboptimality decays from 6.5×10^{-2} , 1.8×10^{-2} and 5.3×10^{-3} at $N = 100$ to 2.4×10^{-4} , 4.8×10^{-5} and 4.1×10^{-6} at $N = 3 \times 10^4$. The slope is close to

Table 2: E3: relative suboptimality (median / worst over 40 tasks).

M	best library policy	SF-GPI	GPI-init + 1 step
1	0.220 / 1.460	0.0071 / 0.377	0.0071 / 0.377
2	0.131 / 1.218	0.0047 / 0.258	0.0055 / 0.261
4	0.079 / 0.519	0.0019 / 0.056	0.0021 / 0.077
8	0.063 / 0.311	0.0014 / 0.037	0.0014 / 0.080
16	0.041 / 0.142	0.0007 / 0.013	0.0011 / 0.030

−1, which matches the quadratic regime discussed after Theorem 12.

The certificate of Theorem 12(a) is sound but conservative. It never failed when issued. With the true error $\|\hat{\Theta} - \Theta\|$ in place of $\bar{\epsilon}_N$, however, it covered only 3%, 9% and 28% of the tasks at $N = 3 \times 10^4$, and with $\bar{\epsilon}_N$ it covered none. The factor $\|\hat{P}\|/c(W)$ causes this conservatism; it ranges from 23 for $Q = I, R = I$ to more than 10^3 for the most ill-conditioned random tasks. For $Q = I, R = I$ and $\sigma_e = 3$, the certificate with $\bar{\epsilon}_N$ would require $N \gtrsim 1.7 \times 10^5$. Section 7 discusses how to tighten the constants.

6.4. E3: Zero-Iteration SF-GPI

With a noise-free representation ($N = 1000$), we built libraries of $M \in \{1, 2, 4, 8, 16\}$ gains, each optimal for a random anchor task, and solved 40 new test tasks in Mode GPI. We obtained the closed-loop cost of the switched policy (15) by simulation from 10 initial states. Fig. 4 and Table 2 compare SF-GPI with the best single library policy and with one Newton step from the library initialization. In median suboptimality, GPI improves on the best library policy by a factor of 30–60 and reaches 0.07% with $M = 16$. Its test-time cost is 0.03–0.12 ms, and it solves no equation. Every GPI closed loop was stable, as Theorem 9 guarantees.

6.5. E4: Continuous Time

We excited the CT reactor with $u = -K_b x + e(t)$ over 2 s, split into $l = 100$ intervals. Here K_b is one stabilizing gain, and e is a sum of ten sinusoids per channel. The rank condition holds ($18 = 18$), and the Gram representation (21) is 34×34 . We solved 50 random tasks from the same $K_0 = K_b$. Algorithm 3 converged in 6.8 iterations on average, and all iterates were stabilizing. The gains agree with per-task off-policy ADP Jiang and Jiang (2012) to 8×10^{-11} . Both sets of gains differ from the CARE solution by a relative error of 2×10^{-4} (Fig. 5(a)). This error comes from the numerical quadrature of the integrals, which both methods share. Fig. 5(b) shows closed-loop responses for three very different tasks computed from the same data.

6.6. E5: Tasks Specified by Reward Labels

Instead of (Q, R) , the learner received labels $r_k = z_k^\top W z_k + v_k$ on the $N = 1000$ stored samples. The noise v_k is Gaussian, with standard deviation equal to a fraction of $\text{std}(r)$. Table 3 shows that noise-free labels identify W exactly. With 1% label noise, the weight error is 2% and the median suboptimality is 5×10^{-5} , which indicates that the LQR solution is insensitive to moderate weight errors. At this noise level, reward labels are a practical way to specify tasks.

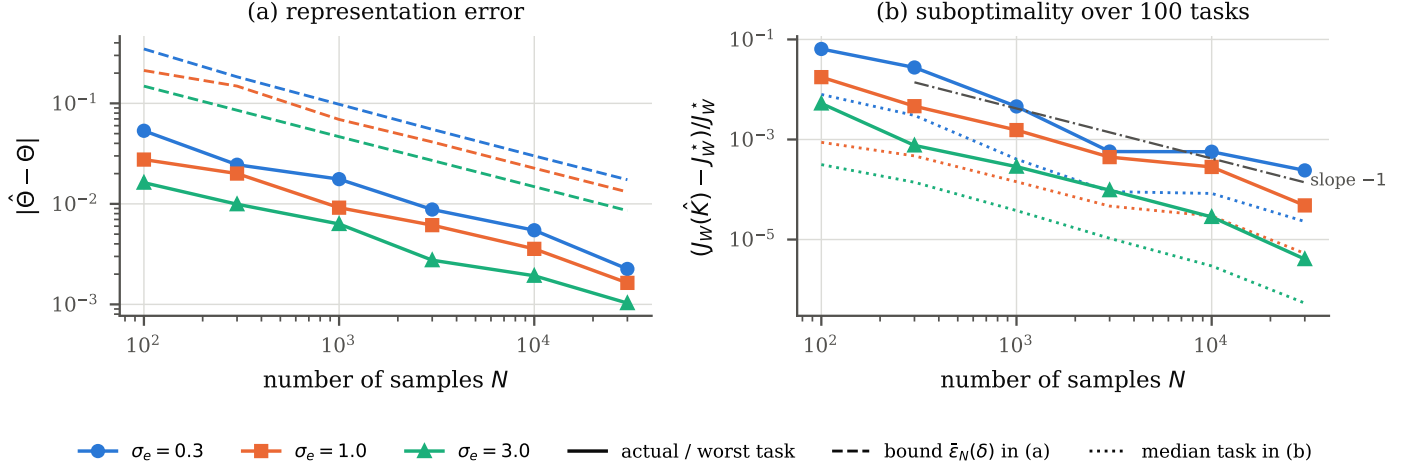


Figure 3: E2: (a) representation error versus N for three excitation levels (solid: observed, dashed: bound $\bar{\epsilon}_N(\delta)$ of Lemma 10); (b) worst-task (solid) and median-task (dotted) relative suboptimality over 100 tasks.

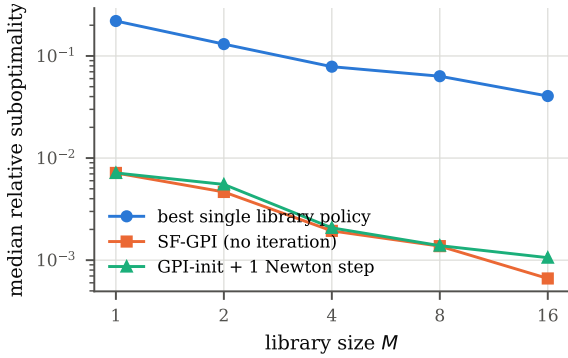


Figure 4: E3: median relative suboptimality over 40 unseen tasks versus library size.

Table 3: E5: task inference from noisy reward labels (50 tasks).

label noise	median $\frac{\ \hat{W} - W\ _F}{\ W\ _F}$	median subopt.	worst subopt.
0	1.7×10^{-14}	≈ 0	3.3×10^{-14}
1%	0.020	4.6×10^{-5}	3.0×10^{-3}
10%	0.204	5.5×10^{-3}	0.168
100%	2.12	0.191	1.76

7. Discussion and Limitations

Relation to certainty equivalence. In the linear-quadratic case, the zero-shot representation is equivalent to the least-squares estimate of Θ , and Mode PI returns the certainty-equivalent gain. This equivalence follows from the structure of the problem: by Proposition 6(c), every successor feature depends on the system only through Θ . The zero-shot viewpoint adds three elements: (i) the task-linear successor operator, which makes policy evaluation for a new task a matrix-vector product and enables zero-iteration GPI; (ii) guarantees that hold for all tasks simultaneously, with a per-task certificate computed from data; and (iii) a template for the nonlinear case, where no closed-form model exists.

Conservatism. The constants in Theorem 12 are explicit but loose. The looseness comes mainly from the factor $\|\hat{P}\|/c(W)$ and from the linear dependence on $\bar{\epsilon}_N$. Three tools could sharpen the certificate. They are the quadratic perturbation theory of Mania et al. (2019), self-normalized bounds in the ℓ_2 norm Simchowitz et al. (2018) in place of per-row union bounds, and robust synthesis over the confidence set Dean et al. (2020).

Assumptions. We assume full-state measurement, $Q > 0$, and, for the certificate, a known noise level σ_w . Natural next steps are extensions to detectable $(A, Q^{1/2})$, to output feedback Rizvi and Lin (2019), and to H_∞ or tracking tasks. In H_∞ and tracking problems, the task also contains an attenuation level or a reference signal, respectively.

Nonlinear systems. For $\dot{x} = f(x, u)$, the same two-phase structure applies with a lifted feature map ϕ (e.g., Koopman observables Korda and Mezić (2018) or neural features) and costs linear in ϕ . The successor operator then becomes approximate, and the analysis requires ADP-type arguments that establish uniform ultimate boundedness (UUB). We regard this nonlinear extension as the main open direction.

8. Conclusion

We formulated zero-shot reinforcement learning for linear quadratic regulation and showed that, in this setting, successor features and the forward-backward factorization become exact and finite-dimensional. The resulting algorithm learns once from reward-free, off-policy data. For any quadratic performance index revealed at test time, it returns a stabilizing gain. The suboptimality of this gain is bounded uniformly over tasks in terms of the amount of data and the PE level. Alternatively, it returns a zero-iteration GPI controller with a Lyapunov stability guarantee. Simulations on the unstable batch reactor confirm exactness with noise-free data, a test-time cost independent of data length, the predicted sample complexity, and the near-optimality of SF-GPI.

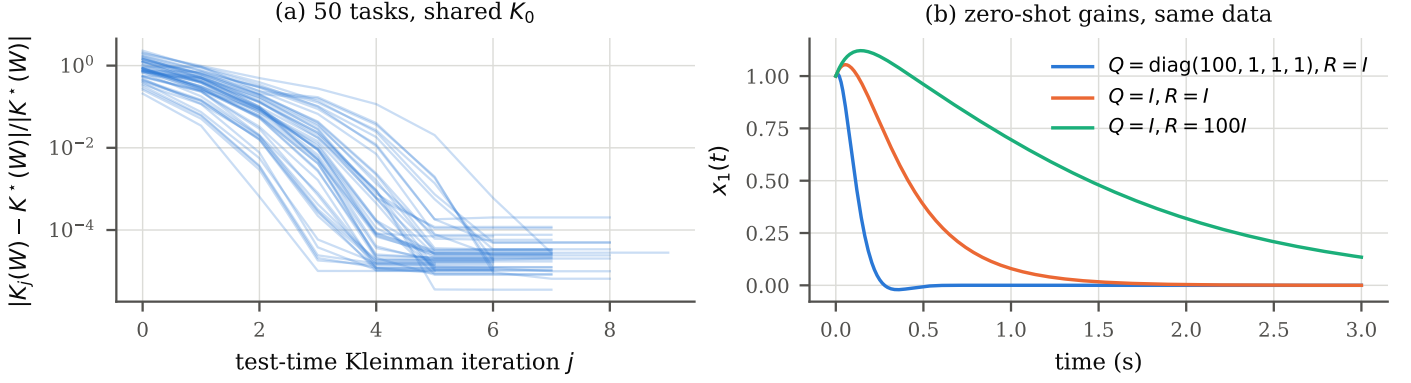


Figure 5: E4 (CT): (a) data-free Kleinman iteration for 50 tasks from one stabilizing gain; (b) first state of the closed loop under zero-shot gains for three tasks.

CRedit authorship contribution statement

Huy Dao: Conceptualization, Methodology, Software, Writing – original draft. **Khanh Thai:** Conceptualization, Methodology, Software, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

[TODO: funding information]

Appendix A. Proofs

This appendix proves every result of the paper step by step. Section [Appendix A.1](#) collects eight elementary facts, and each later step names the fact it uses.

Appendix A.1. Elementary Facts

Throughout, $X \leq Y$ means that $Y - X$ is positive semidefinite (the Loewner order).

(F1) Loewner order. For symmetric X, Y and any matrix T of compatible size:

- (i) $X \leq Y \Rightarrow T^\top X T \leq T^\top Y T$, because $v^\top T^\top (Y - X) T v = (Tv)^\top (Y - X) (Tv) \geq 0$;
- (ii) $\lambda_{\min}(X)I \leq X \leq \lambda_{\max}(X)I$, and $\|X\| = \lambda_{\max}(X)$ if $X \geq 0$;
- (iii) if $0 < X \leq Y$ then $Y^{-1} \leq X^{-1}$;
- (iv) if $0 \leq X \leq Y$ then $\|X\| \leq \|Y\|$ and $\text{tr } X \leq \text{tr } Y$;
- (v) if $-aI \leq X \leq bI$ with $a, b \geq 0$ then $\|X\| \leq \max\{a, b\}$.

(F2) Discrete Lyapunov equation. Let F be Schur and D symmetric. Then $X = F^\top X F + D$ has the unique solution

$$X = \sum_{t=0}^{\infty} (F^\top)^t D F^t,$$

and $D \geq 0$ implies $X \geq 0$.

Proof. Substituting the equation into itself T times gives

$$\begin{aligned} X &= F^\top X F + D \\ &= F^\top (F^\top X F + D) F + D \\ &= (F^\top)^2 X F^2 + F^\top D F + D \\ &\vdots \\ &= (F^\top)^T X F^T + \sum_{t=0}^{T-1} (F^\top)^t D F^t. \end{aligned}$$

Since F is Schur, $F^T \rightarrow 0$, so the first term vanishes as $T \rightarrow \infty$ and the series converges. If X' is a second solution, then $Y = X - X'$ satisfies $Y = F^\top Y F$, so $Y = (F^\top)^T Y F^T \rightarrow 0$ and $Y = 0$. Each term of the series is ≥ 0 when $D \geq 0$ by (F1)(i). $\square$

(F3) Comparison principle. Let F be Schur, $X = F^\top X F + D$, and let Y be symmetric with $F^\top Y F + D \leq Y$. Then $X \leq Y$.

Proof. Subtracting the two relations gives

$$Y - X = F^\top (Y - X) F + \underbrace{(Y - F^\top Y F - D)}_{\geq 0}.$$

By (F2), $Y - X = \sum_t (F^\top)^t (Y - F^\top Y F - D) F^t \geq 0$. $\square$

(F4) Lyapunov stability test. If $P > 0$ and $F^\top P F - P < 0$, then F is Schur.

Proof. Let $Fv = \lambda v$ with $v \in \mathbb{C}^n$, $v \neq 0$. Then

$$v^* (F^\top P F - P) v = (|\lambda|^2 - 1) v^* P v.$$

The left-hand side is negative and $v^* P v > 0$, so $|\lambda| < 1$. $\square$

(F5) Kronecker identities.

- (i) $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$;
- (ii) $(X \otimes Y)(U \otimes V) = (XU) \otimes (YV)$;
- (iii) the eigenvalues of $X \otimes Y$ are the products $\lambda_i(X)\lambda_j(Y)$, so $\rho(X \otimes Y) = \rho(X)\rho(Y)$;
- (iv) $z^\top W z = \text{vec}(W)^\top (z \otimes z)$, which is (i) with $X = z^\top$, $Y = W$, $Z = z$.

(F6) XY and YX . For $X \in \mathbb{R}^{d \times n}$ and $Y \in \mathbb{R}^{n \times d}$, XY and YX have the same nonzero eigenvalues.

Proof. Let $XYv = \lambda v$ with $\lambda \neq 0$, $v \neq 0$. Then $Yv \neq 0$, since otherwise $\lambda v = XYv = 0$. The vector Yv satisfies $YX(Yv) = Y(XYv) = \lambda Yv$, so λ is an eigenvalue of YX . The converse is symmetric. $\square$

(F7) Minimizing a quadratic in u . Let $H = \begin{bmatrix} H^{xx} & H^{xu} \\ H^{ux} & H^{uu} \end{bmatrix}$ be symmetric with $H^{uu} > 0$, and let $K^+ := (H^{uu})^{-1}H^{ux}$. Then for all x, u

$$\begin{bmatrix} x \\ u \end{bmatrix}^\top H \begin{bmatrix} x \\ u \end{bmatrix} = (u + K^+x)^\top H^{uu}(u + K^+x) + x^\top (H^{xx} - H^{xu}K^+)x,$$

so $\min_u [x; u]^\top H [x; u] = x^\top (H^{xx} - H^{xu}K^+)x$, attained only at $u = -K^+x$.

Proof. Expand both sides and use $H^{uu}K^+ = H^{ux}$:

$$\begin{aligned} & (u + K^+x)^\top H^{uu}(u + K^+x) \\ &= u^\top H^{uu}u + 2u^\top H^{uu}K^+x + x^\top K^{+\top} H^{uu}K^+x \\ &= u^\top H^{uu}u + 2u^\top H^{ux}x + x^\top H^{xu}K^+x. \end{aligned}$$

Adding $x^\top H^{xx}x - x^\top H^{xu}K^+x$ gives $x^\top H^{xx}x + 2u^\top H^{ux}x + u^\top H^{uu}u = [x; u]^\top H [x; u]$. $\square$

(F8) Matrix Young inequality. For $P \geq 0$ and $\eta > 0$,

$$(X + E)^\top P(X + E) \leq (1 + \eta)X^\top PX + (1 + \eta^{-1})E^\top PE.$$

Proof. By (F1)(i),

$$\begin{aligned} 0 &\leq (\sqrt{\eta}X - E/\sqrt{\eta})^\top P(\sqrt{\eta}X - E/\sqrt{\eta}) \\ &= \eta X^\top PX - X^\top PE - E^\top PX + \eta^{-1}E^\top PE, \end{aligned}$$

so $X^\top PE + E^\top PX \leq \eta X^\top PX + \eta^{-1}E^\top PE$. Insert this bound on the cross terms into

$$(X + E)^\top P(X + E) = X^\top PX + X^\top PE + E^\top PX + E^\top PE. \quad \square$$

Appendix A.2. Proof of Lemma 5

Write $\Pi = \Pi_K$, $G = G_K$, $P = P_K(W)$, $H = H_K(W)$.

Step 1 (spectral radius). Apply (F6) with $X = \Pi \in \mathbb{R}^{d \times n}$ and $Y = \Theta \in \mathbb{R}^{n \times d}$:

$$XY = \Pi\Theta = G, \quad YX = \Theta\Pi = A - BK.$$

The two matrices share their nonzero eigenvalues, so $\rho(G) = \rho(A - BK)$. In particular G is Schur if and only if $K \in \mathcal{K}(\Theta)$.

Step 2 (stage cost as a quadratic form). Since $W = \text{blkdiag}(Q, R)$,

$$\Pi^\top W \Pi = \begin{bmatrix} I & -K^\top \end{bmatrix} \begin{bmatrix} Q & 0 \\ 0 & R \end{bmatrix} \begin{bmatrix} I \\ -K \end{bmatrix} = Q + K^\top RK.$$

Step 3 ($P = \Pi^\top H \Pi$). Using $A - BK = \Theta\Pi$ and Step 2, the Lyapunov equation (3) becomes

$$P = (\Theta\Pi)^\top P(\Theta\Pi) + \Pi^\top W \Pi$$

$$\begin{aligned} &= \Pi^\top (\Theta^\top P \Theta) \Pi + \Pi^\top W \Pi \\ &= \Pi^\top (W + \Theta^\top P \Theta) \Pi \\ &= \Pi^\top H \Pi. \end{aligned}$$

Step 4 (Lyapunov equation for H). Using $G = \Pi\Theta$ and Step 3,

$$\begin{aligned} G^\top H G &= \Theta^\top (\Pi^\top H \Pi) \Theta \\ &= \Theta^\top P \Theta \\ &= H - W, \end{aligned}$$

where the last equality is the definition (6). Hence $H = W + G^\top H G$.

Step 5 (uniqueness). G is Schur by Step 1, so (F2) gives uniqueness. $\square$

Appendix A.3. Proof of Proposition 6

Let $G = G_K$.

Step 1 (trajectory). Start at $z_0 = z = [x; u]$, i.e., take the action u first and follow K afterwards. Then $x_1 = \Theta z_0$ and $u_1 = -Kx_1$, so

$$z_1 = \begin{bmatrix} x_1 \\ -Kx_1 \end{bmatrix} = \Pi_K x_1 = \Pi_K \Theta z_0 = G z_0.$$

The same argument at every later step gives $z_t = G^t z$ by induction.

Step 2 (features along the trajectory). By (F5)(ii),

$$\begin{aligned} \phi(z_t) &= (G^t z) \otimes (G^t z) \\ &= (G^t \otimes G^t)(z \otimes z) \\ &= (G \otimes G)^t(z \otimes z). \end{aligned}$$

Step 3 (summing the series, part (a)). By (F5)(iii) and Lemma 5, $\rho(G \otimes G) = \rho(G)^2 < 1$. The Neumann series therefore converges:

$$\begin{aligned} \psi^K(z) &= \sum_{t \geq 0} \phi(z_t) = \left(\sum_{t \geq 0} (G \otimes G)^t \right) (z \otimes z) \\ &= (I - G \otimes G)^{-1}(z \otimes z). \end{aligned}$$

Each entry of $\psi^K(z)$ is a linear combination of the entries of $z \otimes z$, i.e., a quadratic form in z .

Step 4 (Q-function, part (b) first claim). The Q-function is the cost of the whole trajectory of Step 1. By (F5)(iv),

$$\begin{aligned} Q_K^W(z) &= \sum_{t \geq 0} z_t^\top W z_t \\ &= \sum_{t \geq 0} \text{vec}(W)^\top \phi(z_t) \\ &= \psi^K(z)^\top \text{vec}(W). \end{aligned}$$

Step 5 (successor operator, part (b) second claim). Vectorize $H_K = W + G^\top H_K G$ (Lemma 5) with (F5)(i), taking $X = G^\top$, $Y = H_K$, $Z = G$:

$$\text{vec}(H_K) = \text{vec}(W) + (G^\top \otimes G^\top) \text{vec}(H_K).$$

Hence $(I - G^\top \otimes G^\top) \text{vec}(H_K) = \text{vec}(W)$. The matrix on the left is invertible because $\rho(G^\top \otimes G^\top) = \rho(G)^2 < 1$, which gives $\text{vec}(H_K) = \mathcal{L}_K \text{vec}(W)$.

Step 6 (part (c)). $G = \Pi_K \Theta$ contains the system only through Θ , and so do $\mathcal{L}_K$ and ψ^K . $\square$

Appendix A.4. Proof of Theorem 8

Step 1 ($\hat{\Theta} = \Theta$). With $w_k \equiv 0$, every sample satisfies $x_{k+1} = \Theta z_k$. Stacking the samples gives $X^+ = \Theta Z$. PE means $ZZ^\top$ is invertible, so

$$\begin{aligned}\hat{\Theta} &= X^+ Z^\top (ZZ^\top)^{-1} \\ &= \Theta ZZ^\top (ZZ^\top)^{-1} \\ &= \Theta.\end{aligned}$$

Consequently $\hat{G}_K = G_K$, $\hat{H}_K = H_K$ and $\hat{P}_K = P_K$ for every K and every W . This step does not involve W .

Step 2 (the iteration is Hewer's algorithm). Suppose $K_j \in \mathcal{K}(\Theta)$; Step 4 below shows this holds for every j . By Lemma 5, the first equation in (14) returns $H_{K_j}(W)$. With $\Theta = [A \ B]$ and $P_j := P_{K_j}(W)$,

$$\Theta^\top P_j \Theta = \begin{bmatrix} A^\top P_j A & A^\top P_j B \\ B^\top P_j A & B^\top P_j B \end{bmatrix},$$

so the blocks of $H_{K_j} = W + \Theta^\top P_j \Theta$ are

$$H^{uu} = R + B^\top P_j B, \quad H^{ux} = B^\top P_j A.$$

The second equation in (14) therefore reads

$$K_{j+1} = (R + B^\top P_j B)^{-1} B^\top P_j A,$$

which is exactly Hewer's update [Hewer \(1971\)](#). Below, $S_j := R + B^\top P_j B > 0$, so that

$$B^\top P_j A = S_j K_{j+1}. \quad (\text{A.1})$$

Step 3 (a key identity). For an arbitrary gain L define

$$f(L) := (A - BL)^\top P_j (A - BL) + Q + L^\top RL.$$

Expanding the product term by term:

$$\begin{aligned}f(L) &= A^\top P_j A - A^\top P_j BL - L^\top B^\top P_j A \\ &\quad + L^\top B^\top P_j BL + Q + L^\top RL \\ &= A^\top P_j A + Q - A^\top P_j BL - L^\top B^\top P_j A \\ &\quad + L^\top S_j L.\end{aligned}$$

Substituting (A.1), i.e. $B^\top P_j A = S_j K_{j+1}$ and $A^\top P_j B = K_{j+1}^\top S_j$:

$$f(L) = A^\top P_j A + Q - K_{j+1}^\top S_j L - L^\top S_j K_{j+1} + L^\top S_j L.$$

Completing the square, using

$$\begin{aligned}(L - K_{j+1})^\top S_j (L - K_{j+1}) \\ = L^\top S_j L - L^\top S_j K_{j+1} - K_{j+1}^\top S_j L + K_{j+1}^\top S_j K_{j+1},\end{aligned}$$

we obtain for every L :

$$\begin{aligned}f(L) &= A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1} \\ &\quad + (L - K_{j+1})^\top S_j (L - K_{j+1}).\end{aligned} \quad (\text{A.2})$$

Now evaluate (A.2) at two gains.

- $L = K_j$: $f(K_j) = P_j$ is the Lyapunov equation of P_j . Hence

$$P_j = A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1} + \Delta_j^\top S_j \Delta_j,$$

with $\Delta_j := K_j - K_{j+1}$.

- $L = K_{j+1}$: the last term vanishes, so

$$f(K_{j+1}) = A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1}.$$

Subtracting the second line from the first, and writing $A_{j+1} := A - BK_{j+1}$:

$$\begin{aligned}P_j &= A_{j+1}^\top P_j A_{j+1} + Q + K_{j+1}^\top R K_{j+1} \\ &\quad + \Delta_j^\top S_j \Delta_j.\end{aligned} \quad (\text{A.3})$$

Step 4 (part (a): every iterate is stabilizing). We argue by induction on j ; $K_0 \in \mathcal{K}(\Theta)$ by assumption. Assume $K_j \in \mathcal{K}(\Theta)$. Then $P_j \geq Q > 0$ by (F2). Write (A.3) as

$$A_{j+1}^\top P_j A_{j+1} - P_j = -(Q + K_{j+1}^\top R K_{j+1} + \Delta_j^\top S_j \Delta_j).$$

The right-hand side is $\leq -Q < 0$. By (F4), A_{j+1} is Schur, so $K_{j+1} \in \mathcal{K}(\Theta)$. In words: *the value matrix of the old gain is a Lyapunov function for the new gain.*

Step 5 (part (b): monotonicity). Since K_{j+1} is stabilizing, P_{j+1} solves

$$P_{j+1} = A_{j+1}^\top P_{j+1} A_{j+1} + Q + K_{j+1}^\top R K_{j+1}.$$

Subtracting this from (A.3):

$$P_j - P_{j+1} = A_{j+1}^\top (P_j - P_{j+1}) A_{j+1} + \Delta_j^\top S_j \Delta_j.$$

Since A_{j+1} is Schur and $\Delta_j^\top S_j \Delta_j \geq 0$, (F2) gives $P_j - P_{j+1} \geq 0$. The lower bound $P^* \leq P_{j+1}$ holds because, for every x_0 , $x_0^\top P^* x_0$ is the smallest achievable cost and $x_0^\top P_{j+1} x_0$ is the cost of one particular policy.

Step 6 (part (c): convergence). For every fixed v , the scalar sequence $v^\top P_j v$ is nonincreasing (Step 5) and bounded below by $v^\top P^* v$, so it converges. Taking $v = e_i$ and $v = e_i + e_l$ shows that every entry of P_j converges; let $\bar{P} := \lim_j P_j$. By continuity,

$$K_j \rightarrow \bar{K} := (R + B^\top \bar{P} B)^{-1} B^\top \bar{P} A.$$

Passing to the limit in (A.3), where $\Delta_j \rightarrow 0$, gives $\bar{P} = (A - B\bar{K})^\top \bar{P} (A - B\bar{K}) + Q + \bar{K}^\top R \bar{K}$. Since $\bar{K}$ is the improvement of $\bar{P}$, identity (A.2) written with $\bar{P}$ in place of P_j and evaluated at $L = \bar{K}$ turns the right-hand side into

$$\begin{aligned}\bar{P} &= A^\top \bar{P} A + Q - \bar{K}^\top (R + B^\top \bar{P} B) \bar{K} \\ &= A^\top \bar{P} A + Q - A^\top \bar{P} B (R + B^\top \bar{P} B)^{-1} B^\top \bar{P} A,\end{aligned}$$

which is the DARE (5). Because $Q > 0$, $(A, Q^{1/2})$ is observable, and the DARE has exactly one positive semidefinite solution, namely P^* [Lancaster and Rodman \(1995\)](#). Since $\bar{P} \geq P^* \geq 0$, it follows that $\bar{P} = P^*$ and $\bar{K} = K^*$. Hewer's iteration is Newton's method applied to the Riccati operator, which gives the quadratic rate [Hewer \(1971\)](#); [Lancaster and Rodman \(1995\)](#).

Step 7 (equivalence with off-policy Q-learning). To evaluate K , off-policy Q-learning [Bradtke et al. \(1994\)](#); [Kiumarsi et al. \(2014\)](#) solves in the least-squares sense the N equations

$$z_k^\top H z_k - \tilde{z}_{k+1}^\top H \tilde{z}_{k+1} = z_k^\top W z_k, \quad \tilde{z}_{k+1} := \Pi_K x_{k+1}.$$

Without noise, $\tilde{z}_{k+1} = \Pi_K \Theta z_k = G_K z_k$, so the k -th equation reads

$$z_k^\top (H - G_K^\top H G_K - W) z_k = 0. \quad (\text{A.4})$$

- H_K is a solution. By Lemma 5, $H_K - G_K^\top H_K G_K - W = 0$, so every equation holds with zero residual.
- The solution is unique. Let H' be another symmetric solution, set $X := H' - H_K$, and let $Y := X - G_K^\top X G_K$ (symmetric). Subtracting (A.4) for H' and for H_K gives $z_k^\top Y z_k = 0$, i.e. $\langle Y, z_k z_k^\top \rangle = 0$ for all k , in the trace inner product. If the matrices $z_k z_k^\top$ span the symmetric matrices, then Y is orthogonal to every symmetric matrix, including itself, so $\langle Y, Y \rangle = 0$ and $Y = 0$. Thus $X = G_K^\top X G_K$, and iterating gives $X = (G_K^\top)^i X G_K^i \rightarrow 0$ because G_K is Schur. Hence $H' = H_K$.

Both methods therefore evaluate the same $H_{K_i}(W)$ and apply the same improvement formula. Starting from the same K_0 , their iterates coincide by induction.

Step 8 (backward map). Noise-free labels satisfy $r_k = z_k^\top W z_k = \phi(z_k)^\top \text{vec}(W)$. The pseudo-inverse in (12) returns the minimum-norm least-squares solution, which lies in the span of the vectors $\phi(z_k) = \text{vec}(z_k z_k^\top)$, so $\hat{w} = \text{vec}(\hat{W})$ with $\hat{W}$ symmetric. The residual is zero because w itself fits every label, hence $\langle \hat{W} - W, z_k z_k^\top \rangle = 0$ for all k . The spanning argument of Step 7 then gives $\hat{W} = W$. $\square$

Appendix A.5. Proof of Theorem 9

Write $Q_i := Q_{K_i}^W$, $P_i := P_{K_i}(W)$, $H_i := H_{K_i}(W)$ and $\ell(x, u) := x^\top Q x + u^\top R u$.

Step 1 (the library Q-functions are exact). Since $\hat{\Theta} = \Theta$ and $K_i \in \mathcal{K}(\Theta)$, Lemma 5 and Proposition 6 show that $\hat{\mathcal{L}}_{K_i} \text{vec}(W)$ returns the true kernel H_i . By Step 2 of the proof of Theorem 8, $H_i^{\text{true}} = R + B^\top P_i B > 0$.

Step 2 (closed form of GPI). By (F7) applied to H_i ,

$$\min_u Q_i(x, u) = x^\top S_i x,$$

with unique minimizer $u = -K_i^+ x$. Minimizing over i as well,

$$V(x) = \min_i x^\top S_i x = \min_i \min_u Q_i(x, u).$$

For $i = i^*(x)$ and $u = \pi_W^{\text{GPI}}(x) = -K_i^+ x$, both minima are attained, so

$$V(x) = Q_i(x, u). \quad (\text{A.5})$$

Step 3 (two Bellman identities). For every i and (x, u) , with $x^+ = Ax + Bu = \Theta z$:

$$\begin{aligned} Q_i(x, u) &= z^\top (W + \Theta^\top P_i \Theta) z \\ &= \ell(x, u) + x^{+\top} P_i x^+, \end{aligned} \quad (\text{A.6})$$

$$\begin{aligned} x^{+\top} P_i x^+ &= x^{+\top} \Pi_{K_i}^\top H_i \Pi_{K_i} x^+ \\ &= Q_i(x^+, -K_i x^+), \end{aligned} \quad (\text{A.7})$$

where (A.7) uses $P_i = \Pi_{K_i}^\top H_i \Pi_{K_i}$ (Lemma 5).

Step 4 (one-step decrease). Fix x , and let $i = i^*(x)$, $u = \pi_W^{\text{GPI}}(x)$, $x^+ = Ax + Bu$. Then

$$\begin{aligned} V(x) &= Q_i(x, u) && \text{by (A.5)} \\ &= \ell(x, u) + x^{+\top} P_i x^+ && \text{by (A.6)} \\ &= \ell(x, u) + Q_i(x^+, -K_i x^+) && \text{by (A.7)} \\ &\geq \ell(x, u) + \min_{u'} Q_i(x^+, u') && \min \leq \text{any value} \\ &= \ell(x, u) + x^{+\top} S_i x^+ && \text{by Step 2} \\ &\geq \ell(x, u) + V(x^+) && V \text{ is a min over } i. \end{aligned}$$

Therefore

$$V(x^+) \leq V(x) - \ell(x, u) \leq V(x) - x^\top Q x. \quad (\text{A.8})$$

The two inequalities are the two ways in which GPI does better than committing to K_i forever: it may re-choose the action (4th line) and the library policy (6th line) at the next step.

Step 5 (quadratic bounds on V). Lower bound. By (A.6) and Step 2,

$$x^\top S_i x = \min_u \{ \ell(x, u) + (Ax + Bu)^\top P_i (Ax + Bu) \} \geq x^\top Q x,$$

because the two terms not involving Q are nonnegative. Hence $S_i \geq Q$ and $V(x) \geq \lambda_{\min}(Q) \|x\|^2$. *Upper bound.* $V(x) \leq x^\top S_1 x \leq \bar{s} \|x\|^2$ with $\bar{s} := \max_i \|S_i\|$.

Step 6 (part (a): exponential stability). From (A.8), Step 5, and $\|x\|^2 \geq V(x)/\bar{s}$:

$$\begin{aligned} V(x^+) &\leq V(x) - \lambda_{\min}(Q) \|x\|^2 \\ &\leq V(x) - \frac{\lambda_{\min}(Q)}{\bar{s}} V(x) \\ &= \kappa V(x), \quad \kappa := 1 - \frac{\lambda_{\min}(Q)}{\bar{s}}. \end{aligned}$$

Since $S_i \geq Q$, $\bar{s} \geq \lambda_{\min}(Q)$, so $\kappa \in [0, 1)$. Iterating gives $V(x_t) \leq \kappa^t V(x_0)$, and by Step 5

$$\|x_t\|^2 \leq \frac{V(x_t)}{\lambda_{\min}(Q)} \leq \frac{\bar{s}}{\lambda_{\min}(Q)} \kappa^t \|x_0\|^2,$$

which is global exponential stability. Note that V is a minimum of quadratics and hence not smooth; this is harmless, since the discrete-time Lyapunov argument uses only the decrease inequality.

Step 7 (part (b): cost bound). By the first inequality of (A.8), $\ell(x_t, u_t) \leq V(x_t) - V(x_{t+1})$. Summing over $t = 0, \dots, T-1$, the right-hand side telescopes:

$$\sum_{t=0}^{T-1} \ell(x_t, u_t) \leq V(x_0) - V(x_T) \leq V(x_0).$$

Letting $T \rightarrow \infty$ bounds the accumulated cost by $V(x_0)$. Finally, taking $u = -K_i x$ in Step 2 and using (A.7),

$$x^\top S_i x \leq Q_i(x, -K_i x) = x^\top P_i x,$$

so $V(x_0) \leq \min_i x_0^\top P_i x_0$. $\square$

Appendix A.6. Proof of Lemma 10

We use the self-normalized martingale bound of (Abbasi-Yadkori et al., 2011, Theorem 1), restated here in our notation.

(F9) Self-normalized bound. Let $\{\mathcal{F}_k\}$ be a filtration, let $z_k \in \mathbb{R}^d$ be $\mathcal{F}_k$ -measurable, and let the scalar η_k be $\mathcal{F}_{k+1}$ -measurable with $\mathbb{E}[e^{s\eta_k} | \mathcal{F}_k] \leq e^{s^2\sigma^2/2}$ for all $s \in \mathbb{R}$. Let $\Lambda > 0$ be deterministic, $\Gamma := \sum_{k < N} z_k z_k^\top$ and $S := \sum_{k < N} z_k \eta_k$. Then for every $\delta' \in (0, 1)$, with probability at least $1 - \delta'$,

$$S^\top (\Gamma + \Lambda)^{-1} S \leq 2\sigma^2 \log \frac{\det(\Gamma + \Lambda)^{1/2} \det(\Lambda)^{-1/2}}{\delta'}.$$

The name refers to the fact that S is normalized by its own random “covariance” Γ . A plain Hoeffding bound does not apply, because z_k contains x_k , which depends on the past noise; the terms $z_k \eta_k$ are not independent, but they form a martingale.

Step 1 (error formula). With $W_N := [w_0, \dots, w_{N-1}]$, the model (1) gives $X^+ = \Theta Z + W_N$. Hence

$$\begin{aligned} \hat{\Theta} - \Theta &= (\Theta Z + W_N) Z^\top (Z Z^\top)^{-1} - \Theta \\ &= W_N Z^\top (Z Z^\top)^{-1}. \end{aligned}$$

Step 2 (one row at a time). Let $\Gamma := Z Z^\top = N \hat{\Sigma}_{zz}$. Row i of W_N is $(w_{0,i}, \dots, w_{N-1,i})$, so row i of $W_N Z^\top$ is $(\sum_k w_{k,i} z_k)^\top =: S_i^\top$. Therefore

$$\text{row } i \text{ of } (\hat{\Theta} - \Theta) = S_i^\top \Gamma^{-1}, \quad i = 1, \dots, n.$$

Step 3 (apply (F9)). By Assumption 2, z_k is $\mathcal{F}_k$ -measurable and $\eta_k := w_{k,i}$ is conditionally σ_w -sub-Gaussian. Take the deterministic $\Lambda := N\lambda_0 I$ and $\delta' := \delta/n$. With probability at least $1 - \delta/n$,

$$S_i^\top (\Gamma + \Lambda)^{-1} S_i \leq 2\sigma_w^2 \log \frac{\det(\Gamma + \Lambda)^{1/2} \det(\Lambda)^{-1/2}}{\delta/n}. \quad (\text{A.9})$$

Step 4 (remove Λ using PE). On the event $\lambda_0 I \leq \hat{\Sigma}_{zz}$:

$$\begin{aligned} \Gamma &= N \hat{\Sigma}_{zz} \geq N\lambda_0 I = \Lambda \\ \Rightarrow \Gamma + \Lambda &\leq 2\Gamma \\ \Rightarrow \Gamma^{-1} &\leq 2(\Gamma + \Lambda)^{-1} \quad \text{by (F1)(iii)} \\ \Rightarrow S_i^\top \Gamma^{-1} S_i &\leq 2 S_i^\top (\Gamma + \Lambda)^{-1} S_i. \end{aligned}$$

Step 5 (bound the determinant). Let $\mu_1, \dots, \mu_d$ be the eigenvalues of $\hat{\Sigma}_{zz}$; on the event $\hat{\Sigma}_{zz} \leq \bar{z}^2 I$, $\mu_j \leq \bar{z}^2$. The matrices $\Gamma + \Lambda = N(\hat{\Sigma}_{zz} + \lambda_0 I)$ and $\Lambda = N\lambda_0 I$ commute, so

$$\begin{aligned} \frac{\det(\Gamma + \Lambda)}{\det \Lambda} &= \prod_{j=1}^d \frac{N\mu_j + N\lambda_0}{N\lambda_0} \\ &= \prod_{j=1}^d \left(1 + \frac{\mu_j}{\lambda_0}\right) \leq \left(1 + \frac{\bar{z}^2}{\lambda_0}\right)^d. \end{aligned}$$

Taking log and combining with (A.9) and Step 4:

$$S_i^\top \Gamma^{-1} S_i \leq 4\sigma_w^2 \left(\frac{d}{2} \log \left(1 + \frac{\bar{z}^2}{\lambda_0}\right) + \log \frac{n}{\delta} \right). \quad (\text{A.10})$$

Step 6 (from the weighted norm to the row norm). Write $\Gamma^{-1} S_i = \Gamma^{-1/2} (\Gamma^{-1/2} S_i)$. Then

$$\begin{aligned} \|S_i^\top \Gamma^{-1}\|^2 &= (\Gamma^{-1/2} S_i)^\top \Gamma^{-1} (\Gamma^{-1/2} S_i) \\ &\leq \|\Gamma^{-1}\| \|\Gamma^{-1/2} S_i\|^2 \\ &= \frac{S_i^\top \Gamma^{-1} S_i}{\lambda_{\min}(\Gamma)} \leq \frac{S_i^\top \Gamma^{-1} S_i}{N\lambda_0}. \end{aligned}$$

Together with (A.10), each row satisfies

$$\|\text{row } i\|^2 \leq \frac{4\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log \left(1 + \frac{\bar{z}^2}{\lambda_0}\right) + \log \frac{n}{\delta} \right).$$

Step 7 (all rows together). Each of the n bounds (A.9) fails with probability at most δ/n , so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event and on the PE event, summing the n row bounds and using $\|\cdot\| \leq \|\cdot\|_F$:

$$\|\hat{\Theta} - \Theta\|^2 \leq \|\hat{\Theta} - \Theta\|_F^2 = \sum_{i=1}^n \|\text{row } i\|^2 \leq \bar{\epsilon}_N(\delta)^2.$$

Step 8 (the form of the statement). Steps 4–7 show: if the PE event holds and all n bounds (A.9) hold, then $\|\hat{\Theta} - \Theta\| \leq \bar{\epsilon}_N$. Hence the event $\mathcal{E}$ (PE holds but $\|\hat{\Theta} - \Theta\| > \bar{\epsilon}_N$) is contained in the event that some bound (A.9) fails, which has probability at most δ . $\square$

Remark 16. (F9) requires Λ , and hence λ_0 , to be fixed before the data are seen; $\bar{z}$ enters only through the PE event and may be chosen freely. If λ_0 is to be chosen from the data, apply the lemma on the grid $\lambda_0 \in \{2^{-l}\}_{l \geq 1}$ with confidence $\delta_l = \delta/(l(l+1))$. A union bound then covers all grid points at total confidence δ , at the price of an additional $\log(l(l+1))$ inside $\bar{\epsilon}_N$.

Appendix A.7. Proof of Lemma 11

The idea is to use the value matrix P_1 of the nominal model as a Lyapunov function for the perturbed closed loop.

Step 0 (notation). Let $\Pi := \Pi_K$, the one-step cost $M := \Pi^\top W \Pi = Q + K^\top R K$, the closed loops $A_1 := \Theta_1 \Pi$ and $A_2 := \Theta_2 \Pi$, and

$$E := A_2 - A_1 = (\Theta_2 - \Theta_1) \Pi.$$

P_1 solves $P_1 = A_1^\top P_1 A_1 + M$. If $\varepsilon = 0$ there is nothing to prove, so let $\varepsilon > 0$.

Step 1 (four basic inequalities, all expressed in units of M).

- (i) $M \geq c \Pi^\top \Pi \geq cI$. The first inequality is $W \geq cI$ with (F1)(i); the second is $\Pi^\top \Pi = I + K^\top K \geq I$.
- (ii) $p_1 \geq c$, hence $\varepsilon \leq c/(6p_1) \leq 1/6$. Indeed $P_1 = A_1^\top P_1 A_1 + M \geq M \geq cI$, and then (F1)(iv).
- (iii) $E^\top P_1 E \leq \frac{p_1 \varepsilon^2}{c} M$. For every v ,

$$\begin{aligned} v^\top E^\top P_1 E v &\leq p_1 \|(\Theta_2 - \Theta_1) \Pi v\|^2 \\ &\leq p_1 \varepsilon^2 \|\Pi v\|^2 = p_1 \varepsilon^2 v^\top \Pi^\top \Pi v \\ &\leq \frac{p_1 \varepsilon^2}{c} v^\top M v \quad \text{by (i).} \end{aligned}$$

(iv) $P_1 \leq \frac{p_1}{c}M$, because $P_1 \leq p_1 I$ by (F1)(ii) and $I \leq M/c$ by (i).

Step 2 (Young's inequality with $\eta = \varepsilon$). Apply (F8) with $X = A_1$:

$$A_2^\top P_1 A_2 \leq (1 + \varepsilon)A_1^\top P_1 A_1 + (1 + \varepsilon^{-1})E^\top P_1 E.$$

The choice $\eta = \varepsilon$ balances the two prices: $(1 + \eta)$ inflates P_1 by $O(\eta)$, and $(1 + \eta^{-1})$ multiplies $E^\top P_1 E = O(\varepsilon^2)$, giving $O(\varepsilon^2/\eta)$. With $\eta = \varepsilon$ both are $O(\varepsilon)$.

Step 3 (expressing the bound through P_1 and M). Substitute $A_1^\top P_1 A_1 = P_1 - M$ and (iii), noting $(1 + \varepsilon^{-1})\varepsilon^2 = \varepsilon + \varepsilon^2$:

$$\begin{aligned} A_2^\top P_1 A_2 &\leq (1 + \varepsilon)(P_1 - M) + (\varepsilon + \varepsilon^2)\frac{p_1}{c}M \\ &= P_1 + \varepsilon P_1 - (1 + \varepsilon)M + (\varepsilon + \varepsilon^2)\frac{p_1}{c}M. \end{aligned}$$

Bound the term εP_1 with (iv), $\varepsilon P_1 \leq \varepsilon \frac{p_1}{c}M$:

$$\begin{aligned} A_2^\top P_1 A_2 &\leq P_1 - (1 + \varepsilon)M + (2\varepsilon + \varepsilon^2)\frac{p_1}{c}M \\ &= P_1 - \gamma M, \end{aligned}$$

where

$$\gamma := 1 + \varepsilon - (2\varepsilon + \varepsilon^2)\frac{p_1}{c}. \quad (\text{A.11})$$

Step 4 (γ is bounded away from zero). By (ii), $\varepsilon \leq 1/6$, so $\varepsilon^2 \leq \varepsilon/6$ and

$$2\varepsilon + \varepsilon^2 \leq \frac{13}{6}\varepsilon.$$

With $a := \frac{13}{6}\varepsilon \frac{p_1}{c}$, the assumption $\varepsilon \leq c/(6p_1)$ gives $a \leq \frac{13}{36}$. Dropping the term $+\varepsilon \geq 0$ in (A.11),

$$\gamma \geq 1 - a \geq \frac{23}{36} > 0.$$

Step 5 (K stabilizes Θ_2). From Step 3 and (i),

$$A_2^\top P_1 A_2 - P_1 \leq -\gamma M \leq -\gamma c I < 0.$$

Since $P_1 \geq cI > 0$, (F4) shows that A_2 is Schur, i.e. K stabilizes Θ_2 .

Step 6 (cost on Θ_2). Let P_2 be the value matrix on Θ_2 , so $P_2 = A_2^\top P_2 A_2 + M$, and let $Y := P_1/\gamma$. Dividing Step 3 by $\gamma > 0$:

$$A_2^\top Y A_2 \leq Y - M, \quad \text{i.e.} \quad A_2^\top Y A_2 + M \leq Y.$$

Thus Y is a super-solution of the Lyapunov equation of P_2 , and (F3) with $F = A_2$ gives

$$P_2 \leq Y = \frac{1}{\gamma}P_1.$$

Interpretation: the true cost is the sum of the one-step costs M along the trajectory; Y “pays” at least M at every step, so it pays at least the true total.

Step 7 (explicit constant). Using $\gamma \geq 1 - a$ and $a \leq \frac{13}{36}$:

$$\begin{aligned} \frac{1}{\gamma} &\leq \frac{1}{1 - a} = 1 + \frac{a}{1 - a} \leq 1 + \frac{36}{23}a \\ &= 1 + \frac{36}{23} \cdot \frac{13}{6} \cdot \frac{\varepsilon p_1}{c} = 1 + \frac{78}{23} \cdot \frac{\varepsilon p_1}{c} \\ &\leq 1 + 6 \frac{\varepsilon p_1}{c}. \end{aligned}$$

With Step 6 this proves $P_2 \leq (1 + 6\varepsilon p_1/c)P_1$. (The sharper constant $78/23 \approx 3.4$ may be used in place of 6.) $\square$

Appendix A.8. Proof of Theorem 12

Step 0 (one probabilistic event for all tasks). By Lemma 10, with probability at least $1 - \delta$ the following implication holds:

$$\lambda_0 I \leq \hat{\Sigma}_{\mathcal{Z}} \leq \bar{z}^2 I \implies \|\hat{\Theta} - \Theta\| \leq \bar{\varepsilon}_N.$$

From now on we work on this event and assume the PE condition, so $\|\hat{\Theta} - \Theta\| \leq \varepsilon := \bar{\varepsilon}_N$. Everything below is deterministic. Since the event does not depend on W , the conclusions hold for all $W \in \mathcal{W}$ simultaneously; no union bound over tasks is needed. Fix W and write $c = c(W)$, $\hat{p} = \|\hat{P}\|$, $p^* = \|P^*\|$. Mode PI starts from $K_0 \in \mathcal{K}(\hat{\Theta})$, so Hewer's argument (Steps 2–6 of the proof of Theorem 8, which use only the model) applied to $\hat{\Theta}$ shows that its limit $\hat{K}$ stabilizes $\hat{\Theta}$ and that its value on $\hat{\Theta}$ is the stabilizing DARE solution $\hat{P}$ of $\hat{\Theta}$. We write $\ell(x, u) := x^\top Qx + u^\top Ru$.

Part (a). Assume $\varepsilon \leq c/(12\hat{p})$.

Step 1 (from the model to the plant). Apply Lemma 11 with $\Theta_1 = \hat{\Theta}$, $\Theta_2 = \Theta$, $K = \hat{K}$, $P_1 = \hat{P}$. Its condition $\varepsilon \leq c/(6\hat{p})$ holds. Hence $\hat{K} \in \mathcal{K}(\Theta)$ and

$$P_{\hat{K}} \leq (1 + \alpha)\hat{P}, \quad \alpha := \frac{6\varepsilon\hat{p}}{c} \leq \frac{6}{12} = \frac{1}{2}. \quad (\text{A.12})$$

Step 2 (bound p^ by $\hat{p}$).* Optimality of P^* gives $P^* \leq P_{\hat{K}}$. With (A.12) and (F1)(iv),

$$p^* \leq \|P_{\hat{K}}\| \leq (1 + \alpha)\hat{p} \leq \frac{3}{2}\hat{p}.$$

Step 3 (the reverse direction is also allowed). Lemma 11 from Θ to $\hat{\Theta}$ requires $\varepsilon \leq c/(6p^*)$. By Step 2,

$$\frac{c}{6p^*} \geq \frac{c}{6 \cdot \frac{3}{2}\hat{p}} = \frac{c}{9\hat{p}} \geq \frac{c}{12\hat{p}} \geq \varepsilon.$$

Step 4 (from the plant to the model). Apply Lemma 11 with $\Theta_1 = \Theta$, $\Theta_2 = \hat{\Theta}$, $K = K^*$, $P_1 = P^*$. Then K^* stabilizes $\hat{\Theta}$, and its value on $\hat{\Theta}$, denoted $P_{\hat{\Theta}}(K^*)$, satisfies

$$\begin{aligned} P_{\hat{\Theta}}(K^*) &\leq (1 + \beta)P^*, \\ \beta &:= \frac{6\varepsilon p^*}{c} \leq \frac{9\varepsilon\hat{p}}{c} \leq \frac{9}{12} = \frac{3}{4}. \end{aligned}$$

Step 5 (optimality on the model). $\hat{P}$ is optimal for $\hat{\Theta}$, and K^* is one stabilizing gain for $\hat{\Theta}$, so

$$\hat{P} \leq P_{\hat{\Theta}}(K^*) \leq (1 + \beta)P^*.$$

Step 6 (sandwich, first line of (17)). Chaining Step 2, (A.12) and Step 5:

$$P^* \leq P_{\hat{K}} \leq (1 + \alpha)\hat{P} \leq (1 + \alpha)(1 + \beta)P^*.$$

The factor satisfies, using $\beta \leq \frac{3}{4}$ for the product term,

$$\begin{aligned} (1 + \alpha)(1 + \beta) - 1 &= \alpha + \beta + \alpha\beta \\ &\leq \alpha + \beta + \frac{3}{4}\alpha \\ &\leq \left(6 + 9 + \frac{3}{4} \cdot 6\right) \frac{\varepsilon\hat{p}}{c} \end{aligned}$$

$$= 19.5 \frac{\varepsilon \hat{p}}{c} \leq 20 \frac{\varepsilon \hat{p}}{c}.$$

Step 7 (bound on $\|\hat{P} - P^\|$).* From Step 5, $\hat{P} - P^* \leq \beta P^* \leq \beta p^* I$. From Step 6, $P^* - \hat{P} \leq P_{\hat{K}} - \hat{P} \leq \alpha \hat{P} \leq \alpha \hat{p} I$. By (F1)(v),

$$\|\hat{P} - P^*\| \leq \max\{\alpha \hat{p}, \beta p^*\}.$$

The two candidates are $\alpha \hat{p} = 6\varepsilon \hat{p}^2/c$ and $\beta p^* \leq \frac{9\varepsilon \hat{p}}{c} \cdot \frac{3}{2} \hat{p} = 13.5 \varepsilon \hat{p}^2/c$. Hence $\|\hat{P} - P^*\| \leq 14\varepsilon \hat{p}^2/c$.

Step 8 (cost-difference identity). Let $S^* := R + B^\top P^* B$ and $Q^*(x, u) := \ell(x, u) + (Ax + Bu)^\top P^* (Ax + Bu)$, whose kernel is $H^* = W + \Theta^\top P^* \Theta$. By (F7), the DARE, and the formula for K^* ,

$$Q^*(x, u) - x^\top P^* x = (u + K^* x)^\top S^* (u + K^* x). \quad (\text{A.13})$$

Let x_t be the true closed-loop trajectory under $\hat{K}$ from $x_0 = x$, with $u_t = -\hat{K}x_t$. By Step 1 it is stable, so $x_t \rightarrow 0$. Then:

$$\begin{aligned} x^\top P_{\hat{K}} x &= \sum_{t \geq 0} \ell(x_t, u_t), \\ x^\top P^* x &= \sum_{t \geq 0} (x_t^\top P^* x_t - x_{t+1}^\top P^* x_{t+1}), \end{aligned}$$

where the second sum telescopes to $x_0^\top P^* x_0$ because $x_t \rightarrow 0$. Subtracting and using $x_{t+1} = Ax_t + Bu_t$ and (A.13):

$$\begin{aligned} x^\top (P_{\hat{K}} - P^*) x &= \sum_{t \geq 0} (\ell(x_t, u_t) + x_{t+1}^\top P^* x_{t+1} - x_t^\top P^* x_t) \\ &= \sum_{t \geq 0} (Q^*(x_t, u_t) - x_t^\top P^* x_t) \\ &= \sum_{t \geq 0} x_t^\top (K^* - \hat{K})^\top S^* (K^* - \hat{K}) x_t. \end{aligned}$$

Step 9 (bound on $\|\hat{K} - K^\|$).* Every term of the last sum is nonnegative; keeping only $t = 0$ and using $S^* \geq R \geq cI$:

$$x^\top (P_{\hat{K}} - P^*) x \geq c \|\hat{K} - K^*\| x^2.$$

From Step 6, $0 \leq P_{\hat{K}} - P^* \leq 20 \frac{\varepsilon \hat{p}}{c} P^*$, so by (F1)(iv) and Step 2,

$$\|P_{\hat{K}} - P^*\| \leq 20 \frac{\varepsilon \hat{p}}{c} p^* \leq 30 \frac{\varepsilon \hat{p}^2}{c}.$$

Combining the two displays, $c \|\hat{K} - K^*\| x^2 \leq 30 \frac{\varepsilon \hat{p}^2}{c} \|x\|^2$ for all x . Hence $\|\hat{K} - K^*\| \leq 30 \varepsilon \hat{p}^2/c^2$, i.e. $\|\hat{K} - K^*\| \leq \frac{\hat{p}}{c} \sqrt{30\varepsilon}$.

Part (b). Assume $\varepsilon \leq c_W/(18\bar{p}_W)$ and fix $W \in \mathcal{W}$. Then $c \geq c_W$ and $p^* \leq \bar{p}_W$.

Step 10 (the model problem is well posed). Since

$$\varepsilon \leq \frac{c_W}{18\bar{p}_W} \leq \frac{c}{18p^*} \leq \frac{c}{6p^*},$$

Lemma 11 from Θ to $\hat{\Theta}$ with $K = K^*$ applies. Hence K^* stabilizes $\hat{\Theta}$, so the pair $(\hat{A}, \hat{B})$ with $\hat{\Theta} = [\hat{A} \ \hat{B}]$ is stabilizable, and since $Q > 0$ the DARE of $\hat{\Theta}$ has a stabilizing solution $\hat{P}$, to which Mode PI converges (Hewer's argument applied to $\hat{\Theta}$). Moreover

$$\hat{P} \leq P_{\hat{\Theta}}(K^*) \leq \left(1 + \frac{6\varepsilon p^*}{c}\right) P^* \leq \left(1 + \frac{6}{18}\right) P^* = \frac{4}{3} P^*,$$

so $\hat{P} \leq \frac{4}{3} P^* \leq \frac{4}{3} \bar{p}_W$.

Step 11 (the certificate holds automatically). Using Step 10, $c \geq c_W$ and $16 \leq 18$:

$$\frac{c}{12\hat{p}} \geq \frac{c_W}{12 \cdot \frac{4}{3} \bar{p}_W} = \frac{c_W}{16\bar{p}_W} \geq \frac{c_W}{18\bar{p}_W} \geq \varepsilon.$$

Step 12 (uniform rate). By Step 11, part (a) applies to W , so $P_{\hat{K}} \leq (1 + 20\varepsilon \hat{p}/c) P^*$. Taking the trace, which is monotone by (F1)(iv), and using $J_W = \text{tr } P$:

$$\begin{aligned} \frac{J_W(\hat{K}) - J_W^*}{J_W^*} &\leq \frac{20\varepsilon \hat{p}}{c} \leq 20\varepsilon \cdot \frac{\frac{4}{3} \bar{p}_W}{c_W} \\ &= \frac{80}{3} \frac{\bar{p}_W}{c_W} \varepsilon \leq 27 \frac{\bar{p}_W}{c_W} \bar{\varepsilon}_N. \end{aligned}$$

The right-hand side does not depend on W , which gives (18). Substituting the definition (16) of $\bar{\varepsilon}_N$ gives the $O(\cdot)$ form. $\square$

Appendix A.9. Proof of Theorem 15

Let K_j be a gain with $A_j := A - BK_j$ Hurwitz, let $Q_j := Q + K_j^\top RK_j$, let P_j solve $A_j^\top P_j + P_j A_j + Q_j = 0$, and let $K_{j+1} := R^{-1} B^\top P_j$ (Kleinman's step).

Step 1 (the Kleinman pair satisfies (20)). Along the data trajectory, rewrite the dynamics as

$$\dot{x} = A_j x + B(u + K_j x).$$

Differentiating $x^\top P_j x$ and using the Lyapunov equation and $B^\top P_j = RK_{j+1}$:

$$\begin{aligned} \frac{d}{dt}(x^\top P_j x) &= x^\top (A_j^\top P_j + P_j A_j) x + 2(u + K_j x)^\top B^\top P_j x \\ &= -x^\top Q_j x + 2(u + K_j x)^\top RK_{j+1} x. \end{aligned}$$

Integrating over $[t_{i-1}, t_i]$:

$$\begin{aligned} x^\top P_j x \Big|_{t_{i-1}}^{t_i} &= - \int_{t_{i-1}}^{t_i} x^\top Q_j x d\tau \\ &\quad + 2 \int_{t_{i-1}}^{t_i} (u + K_j x)^\top RK_{j+1} x d\tau. \end{aligned} \quad (\text{A.14})$$

Each term is linear in the unknowns. By definition of $\bar{x}$, $x^\top P_j x = \bar{x}^\top \text{svec}(P_j)$. By (F5)(iv), $x^\top Q_j x = (x \otimes x)^\top \text{vec}(Q_j)$. By (F5)(i) with $X = (u + K_j x)^\top R$, $Y = K_{j+1}$, $Z = x$:

$$\begin{aligned} (u + K_j x)^\top RK_{j+1} x &= (x^\top \otimes (u + K_j x)^\top R) \text{vec}(K_{j+1}) \\ &= [(x^\top \otimes x^\top)(I_n \otimes K_j^\top R) + (x^\top \otimes u^\top)(I_n \otimes R)] \text{vec}(K_{j+1}), \end{aligned}$$

where the last line uses (F5)(ii) in the form $x^\top \otimes x^\top K_j^\top R = (x^\top \otimes x^\top)(I_n \otimes K_j^\top R)$, and similarly for u . Substituting into (A.14) and stacking the l intervals gives exactly (20), with zero residual.

Step 2 (the equation depends on data only through D). Let $q := \frac{n(n+1)}{2} + nm$ and

$$T(K_j, R) := \begin{bmatrix} I & 0 \\ 0 & -2(I_n \otimes K_j^\top R) \\ 0 & -2(I_n \otimes R) \end{bmatrix} \in \mathbb{R}^{p \times q},$$

$$s(K_j, W) := \begin{bmatrix} 0 \\ -\text{vec}(Q_j) \\ 0 \end{bmatrix} \in \mathbb{R}^p,$$

with row blocks of sizes $\frac{n(n+1)}{2}$, n^2 and nm matching $D = [\delta_{xx}, I_{xx}, I_{xu}]$. Multiplying out,

$$\begin{aligned} DT &= [\delta_{xx}, -2I_{xx}(I \otimes K_j^\top R) - 2I_{xu}(I \otimes R)], \\ Ds &= -I_{xx} \text{vec}(Q_j), \end{aligned}$$

so (20) is $DT\theta = Ds$, where T and s depend only on (K_j, W) .

Step 3 (normal equations use only the Gram matrix). Multiplying $DT\theta_j^* = Ds$ on the left by $T^\top D^\top$:

$$T^\top (D^\top D) T \theta_j^* = T^\top (D^\top D) s,$$

so the Kleinman pair θ_j^* solves the equation of Algorithm 3, which involves only $\mathcal{R}_{CT} = D^\top D$.

Step 4 (uniqueness). $T^\top D^\top D T$ is invertible if and only if DT has full column rank. We show that $DT\theta = 0$ implies $\theta = 0$, following the argument of Jiang and Jiang (2012). Let $\theta = [\text{svec}(X); \text{vec}(Y)]$ with X symmetric and $DT\theta = 0$. Along the data,

$$\frac{d}{dt}(x^\top X x) = x^\top (A^\top X + X A)x + 2u^\top B^\top X x.$$

Row i of $DT\theta = 0$ therefore reads

$$\begin{aligned} 0 &= \int [x^\top (A^\top X + X A)x + 2u^\top B^\top X x] d\tau \\ &\quad - 2 \int [x^\top K_j^\top R Y x + u^\top R Y x] d\tau \\ &= \int x^\top (A^\top X + X A - K_j^\top R Y - Y^\top R K_j) x d\tau \\ &\quad + 2 \int u^\top (B^\top X - R Y) x d\tau. \end{aligned}$$

The first integral is a row of I_{xx} applied to a symmetric matrix, the second a row of I_{xu} applied to $\text{vec}(B^\top X - RY)$. The rank condition states exactly that these columns are linearly independent, so

$$B^\top X = RY, \quad A^\top X + X A - K_j^\top R Y - Y^\top R K_j = 0.$$

Substituting $RY = B^\top X$ into the second equation gives

$$(A - BK_j)^\top X + X(A - BK_j) = 0.$$

Since $A - BK_j$ is Hurwitz, this Lyapunov equation has only the solution $X = 0$. Then $Y = R^{-1}B^\top X = 0$, so $\theta = 0$.

Step 5 (induction and convergence). By Steps 3–4, if K_j is stabilizing then Algorithm 3 returns exactly Kleinman's K_{j+1} . Kleinman's theorem Kleinman (1968) states that from a stabilizing K_0 every iterate is stabilizing, P_j decreases monotonically, and $K_j \rightarrow K^*(W)$ quadratically. By induction on j the two sequences coincide. The same K_0 serves every W , because stability of $A - BK_0$ does not involve W .

Step 6 (cost). Forming $T^\top \mathcal{R}_{CT} T$ with $T \in \mathbb{R}^{p \times q}$ and solving the $q \times q$ system costs $O(p^2 q + p q^2 + q^3) = O(p^3)$, independent of the number of intervals l . $\square$

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